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Machine Learning for Brain MRI Analysis in Spinal and Bulbar Muscular Atrophy

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Abstract

Spinal and Bulbar Muscular Atrophy is a rare X-linked neurodegenerative disorder primarily affecting lower motor neurons. Although white-matter alterations have been documented, evidence concerning cortical gray-matter involvement remains limited and inconsistent. This thesis investigates whether structural cortical gray-matter differences between patients with SBMA and healthy controls can be identified using complementary supervised and unsupervised machine-learning approaches.

A dataset of 37 T1-weighted structural MRI scans, comprising 20 patients with SBMA and 17 healthy controls, was analyzed using two complementary approaches. In the supervised analysis, GM was extracted using a 1,000-region cortical parcellation and two regional representations were considered: raw regional GM and GM normalized by total GM volume. The ten most informative regions were selected using mutual information within a Leave-One-Out Cross-Validation framework. Four classification algorithms—Support Vector Machine, Logistic Regression, Random Forest, and XGBoost—were evaluated. Explainable artificial intelligence based on SHAP values was subsequently used to characterize the contribution and direction of individual brain regions to the model predictions. In parallel, an unsupervised community detection analysis was performed by constructing subject-specific graphs in which brain parcels were represented as nodes and spatially connected parcels were weighted according to GM similarity. Separate multiplex networks were constructed for the SBMA and healthy control cohorts, and the Leiden algorithm with consensus clustering was used to identify stable communities.

Classification performance was substantially higher when proportional regional GM measures were used compared with raw GM measures. L1-regularized Logistic Regression achieved the best overall balance between sensitivity and specificity, with an accuracy of 0.84 and an F1-score of 0.84. SHAP analyses consistently identified regions of the frontal operculum-insula within the salience/ventral attention network and prefrontal regions of the default mode network as important contributors to the classification. Lower proportional GM values in these regions were generally associated with predictions of SBMA. The community detection analysis revealed

broadly similar organizational structures between the two cohorts, with eight communities identified in healthy controls and seven in SBMA. The resulting partitions showed high agreement, with an Adjusted Rand Index of 0.962 and a Normalized Mutual Information of 0.960. Differences were concentrated in a relatively small number of regions, primarily within the left somatosensory motor network and, to a lesser extent, the dorsal attention network.

Overall, these findings demonstrate the potential of machine learning and network-based approaches to investigate brain involvement in SBMA. The results suggest that localized patterns of regional GM may contain information capable of distinguishing SBMA from healthy controls, including regions outside the primary motor system, while the overall organization of GM similarity remains largely preserved. Given the limited sample size, the identified regions should be validated in larger, independent datasets.

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Contents

Abstract	iii
Acknowledgments	v
1 Introduction	1
1.1 Clinical and Biological Background	1
1.1.1 Genetic Cause	1
1.1.2 Symptoms	2
1.1.3 Diagnosis, Disease Course and Treatment	2
1.2 Previous Research	3
1.3 Contribution of the Present Study	3
2 Computational Background	5
2.1 Machine Learning	5
2.1.1 Supervised vs Unsupervised	5
2.1.2 Cross-Validation	5
2.1.3 Evaluation Metrics	6
2.2 Graphs and Network Analysis	8
2.2.1 Graphs	8
2.2.2 Multiplex Networks	9
2.2.3 Community Detection	10
2.3 Feature Selection	12
2.3.1 Univariate Feature Selection	13
2.4 Classification Algorithms	13
2.4.1 Support Vector Machine	13
2.4.2 Logistic Regression	14
2.4.3 Random Forest and XGBoost	15
2.5 Explainable AI	17
2.5.1 SHAP Values	17

3	Materials and Methods	19
3.1	Dataset	19
3.1.1	Preprocessing	19
3.2	Community Detection Pipeline	20
3.2.1	Graph Construction	20
3.2.2	Multiplex Construction	20
3.2.3	Community Detection	21
3.3	Classification Pipeline	21
3.3.1	Input Data	22
3.3.2	Feature Selection	22
3.3.3	Classification Algorithms	22
3.3.4	Interpretability Analysis	23
4	Results	27
4.1	Community Detection Results	27
4.2	Classification Results	30
5	Discussions and Conclusions	41
5.1	Methodological Recap	41
5.2	Interpretation	42
5.3	Limitation and Future Research	43
A	Community Partitions	45
A.1	Community Partitions for HC	45
A.2	Community Partition for SBMA	50
	References	57

List of Figures

3.1	Overview of the two analytical pipelines.	25
4.1	Consensus community partitions for healthy controls (left) and SBMA patients (right). Colours indicate highest Jaccard matches. HC cluster 7 is shown separately in brown because its match to SBMA cluster 5 is weak.	28
4.2	Jaccard similarity matrix	28
4.3	Models' accuracy and F1.	31
4.4	Models' Confusion Matrices.	31
4.5	Most frequently selected features for GM representation.	32
4.6	Mean Overall SHAP for each model.	36
4.7	Mean Absolute SHAP when selected for each model.	37
4.8	Mean Signed SHAP for each model.	37
4.9	Beeswarm plots showing the SHAP values obtained from the held-out LOOCV predictions using normalized regional GM measures.	39

List of Tables

4.1	Parcels belonging to HC cluster 7.	29
4.2	Parcels whose SBMA cluster differs from the expected best-match cluster in HC.	30
4.3	Most frequently selected features - Raw regional GM.	33
4.4	Most frequently selected features - normalized regional GM.	33
4.5	Top 15 features for each classification model.	34
A.1	Parcel families and parcel numbers in each Healthy Control cluster.	45
A.2	Parcel families and parcel numbers in each SBMA cluster.	51

Chapter 1

Introduction

Spinal and Bulbar Muscular Atrophy (SBMA) is a rare X-linked genetic disorder, with an estimated prevalence of approximately 2.58 per 10,000 individuals in the Veneto region of Italy [1]. It is characterized by progressive neuromuscular degeneration [2] and is also known as Kennedy's disease, after William Kennedy, who first described the disorder in 1968.

1.1 Clinical and Biological Background

1.1.1 Genetic Cause

Spinal and Bulbar Muscular Atrophy is an X-linked recessive disorder that affects males and is caused by a mutation in the androgen receptor (AR) gene located on the X chromosome. The disease is characterized by an abnormal expansion of a CAG trinucleotide repeat in the first exon of the AR gene [3].

Healthy individuals typically have 34 or fewer CAG repeats, whereas 35 repeats are considered to have uncertain clinical significance [4]. Expansions of 36-37 CAG repeats are associated with reduced disease penetrance [5], while expansions of 38 or more repeats are generally considered diagnostic of SBMA [6]. The expanded CAG repeat leads to the production of an androgen receptor protein containing an abnormally long polyglutamine tract, commonly referred to as polyQ-AR. This mutant protein acquires a toxic gain-of-function, which contributes to cellular dysfunction, while simultaneously resulting in a partial loss of normal androgen receptor function [7].

1.1.2 Symptoms

Affected males develop androgen insensitivity and progressive degeneration of motor neurons in the spinal cord and brainstem, hence the terms *spinal* and *bulbar* [8].

Motor Symptoms

Early symptoms consist of muscle cramps and fasciculations, followed by bulbar and proximal limb muscle weakness, resulting in progressive gait impairment. Patients may require wheelchair assistance approximately 2–3 decades after symptom onset [6, 9]. These motor symptoms are common to other motor neuron diseases, such as amyotrophic lateral sclerosis (ALS), which often leads to misdiagnosis [4, 3].

Bulbar involvement causes atrophy of the facial, pharyngeal, and lingual muscles, resulting in dysarthria and dysphagia in the later stages of the disease. Fasciculations are particularly noticeable around the chin and perioral region [9, 4]. As the disease progresses, the risk of aspiration pneumonia increases, which represents the leading cause of death in SBMA patients [6].

Androgen Insensitivity Symptoms

Partial androgen insensitivity results in gynecomastia, characterized by excessive enlargement of male breast tissue. This condition is observed in approximately three quarters of patients and is often one of the earliest clinical signs of the disease. Other symptoms associated with reduced androgen sensitivity include erectile dysfunction, decreased libido, infertility, and testicular atrophy [6, 10, 11].

Metabolic Disturbances

Insulin resistance, hyperlipidemia, fatty liver disease, and abnormal electrocardiographic findings are also common features of SBMA [12, 9].

1.1.3 Diagnosis, Disease Course and Treatment

The diagnosis of SBMA is confirmed through genetic testing identifying an expansion of the CAG trinucleotide repeat on the X chromosome [4].

Symptoms typically begin in adulthood, most commonly between 30 and 50 years of age [9]. The CAG repeat expansion is shown to correlate with the age of onset but not with severity of symptoms [13, 6]. Disease progression is generally slow.

There is currently no curative treatment for SBMA [2]; therefore, management is primarily supportive and aimed at preventing disease-related complications such as falls, fractures, and aspiration, while maintaining patient independence and quality

of life.

1.2 Previous Research

Due to its well-defined genetic origin, SBMA has been extensively investigated at the molecular and biological levels. Given its primary impact on motor neurons, neuroimaging studies have predominantly focused on structural alterations within the motor system. These studies have reported abnormalities in white matter tracts associated with the primary motor cortex, premotor areas, and somatosensory pathways, as well as alterations involving the brainstem and spinal cord [14].

In contrast, relatively few studies have investigated whether SBMA is associated with alterations beyond the motor system, particularly in cortical regions not directly involved in motor function. One of the earliest investigations, a pilot Positron Emission Tomography study, reported frontal hypometabolism and reduced glucose uptake in prefrontal cortical areas [15]. Similarly, a voxel-based morphometry study published in 2007 identified subtle reductions in gray matter volume, primarily in frontal regions [16]. However, these findings have not been consistently replicated in subsequent studies. For instance, Tu et al. [14] reported no significant differences in gray matter volume between SBMA patients and healthy controls, despite observing widespread abnormalities in white matter integrity.

A recent systematic review of cognitive and behavioral involvement in SBMA concluded that the available neuroimaging evidence remains scarce and contradictory [17]. Thus, whether SBMA is associated with cortical GM alterations remains an open question.

1.3 Contribution of the Present Study

To date, only a limited number of studies have directly compared brain MRI data between SBMA patients and healthy controls, and these studies have primarily relied on conventional statistical approaches. In this context, machine learning and explainable artificial intelligence methods may provide a complementary framework for investigating structural differences between the two cohorts.

In this study, we investigate whether and how SBMA is associated with cortical alterations in brain gray matter. In particular, we address the following questions:

- Are there differences in cortical organization between the two cohorts based on gray matter similarity?

- Can SBMA participants be discriminated from healthy controls based on the gray matter characteristics of their brains?
- Which cortical regions contribute most strongly to the classification?

To address the first question, we employ an unsupervised community detection approach, analyzing SBMA and healthy control brains separately to investigate whether the two cohorts exhibit different patterns of brain organization. In particular, graphs are constructed from the imaging data, multiplex networks are built for each cohort, and the Leiden algorithm is applied to identify communities of brain regions with similar organizational patterns.

To address the remaining two questions, we employ four supervised machine learning algorithms—Support Vector Machines (SVM), Logistic Regression, Random Forests, and Extreme Gradient Boosting (XGBoost)—to classify MRI scans from SBMA patients and healthy controls. Explainable artificial intelligence methods are then used to investigate which brain regions contribute most strongly to the model predictions.

To the best of our knowledge, this is the first study to apply machine learning methods to investigate SBMA using brain MRI data.

Through this framework, this thesis aims to investigate whether disease-specific structural alterations extend to cortical regions.

Chapter 2

Computational Background

2.1 Machine Learning

2.1.1 Supervised vs Unsupervised

Supervised and unsupervised learning are the two principal paradigms of machine learning [18]. In supervised learning, a model is trained using labeled data, where each input sample is associated with a known target output. The objective is to learn a mapping from inputs to outputs that can be generalized to unseen data. Examples of supervised learning are classification and regression algorithms.

In contrast, unsupervised learning operates on unlabeled data and aims to identify underlying patterns, structures, or relationships within the data without pre-defined outputs. Common unsupervised tasks include clustering, dimensionality reduction, and anomaly detection.

2.1.2 Cross-Validation

Cross-validation is a resampling technique used to estimate the generalization ability of a machine learning model and to reduce the risk of overfitting [18]. The main idea is to repeatedly divide the available dataset into training and validation subsets, allowing the model performance to be evaluated on data that were not used during training.

Leave-One-Out Cross-Validation

Leave-One-Out Cross-Validation (LOOCV) [19] is a resampling strategy used to estimate the generalization performance of a machine learning model, particularly useful when the available dataset is small. In LOOCV, the model is trained and

evaluated multiple times, with each iteration leaving out a single sample as the test set.

Given a dataset containing N samples, the procedure consists of N iterations. At each iteration, $N - 1$ samples are used for training the model, while the remaining sample is used for testing. The process is repeated until every sample has been used once as the test sample. The final performance estimate is obtained by aggregating the predictions from all folds and calculating the performance metrics across the complete set.

2.1.3 Evaluation Metrics

For community detection analyses, NMI [20] and the ARI were used to quantify agreement between complete partition solutions [21]. The Jaccard index was additionally used to assess the overlap between pairs of individual clusters [22].

Classification performance was assessed using accuracy and the F1-score [23].

NMI

Normalized Mutual Information (NMI) was used to quantify the similarity between two clustering solutions. NMI measures the information shared by two partitions, normalized by their respective entropies:

$$\text{NMI}(U, V) = \frac{2I(U; V)}{H(U) + H(V)},$$

where U and V denote the two partitions, $H(U)$ and $H(V)$ are their entropies, and $I(U; V)$ is their mutual information. Mutual information can be expressed in terms of entropy as

$$I(U; V) = H(U) + H(V) - H(U, V) = \sum_{u \in U} \sum_{v \in V} p(u, v) \log \left(\frac{p(u, v)}{p(u)p(v)} \right),$$

where $H(U, V)$ is the joint entropy of the two partitions. The quantity $p(u, v)$ denotes the joint probability that an observation is assigned to cluster u in partition U and cluster v in partition V , while $p(u)$ and $p(v)$ are the corresponding marginal probabilities. Values range from 0, indicating no agreement, to 1, indicating identical partitions, irrespective of the numerical labels assigned to the clusters.

ARI

The Adjusted Rand Index (ARI) is a measure of clustering agreement. It compares all pairs of elements and counts how often the two clusterings agree: either both

place a pair in the same cluster, or both place the pair in different clusters, corrected for the similarity expected by chance. The Rand index has a value between -0.5 and 1. Random labelings have an ARI close to 0, 1 stands for perfect match and -0.5 indicates especially discordant clusterings. It is defined as

$$\text{ARI}(U, V) = \frac{\sum_{i,j} \binom{n_{ij}}{2} - \frac{\left(\sum_i \binom{a_i}{2}\right) \left(\sum_j \binom{b_j}{2}\right)}{\binom{n}{2}}}{\frac{1}{2} \left[\sum_i \binom{a_i}{2} + \sum_j \binom{b_j}{2} \right] - \frac{\left(\sum_i \binom{a_i}{2}\right) \left(\sum_j \binom{b_j}{2}\right)}{\binom{n}{2}}},$$

where U and V denote the two partitions, n_{ij} is the number of observations assigned to cluster i in U and cluster j in V , $a_i = \sum_j n_{ij}$ and $b_j = \sum_i n_{ij}$ are the corresponding marginal totals, and n is the total number of observations.

Jaccard Index

The Jaccard index was used to quantify the similarity between pairs of individual clusters [22]. In contrast to NMI and ARI, which compare complete clustering partitions, the Jaccard index measures the overlap between two sets of observations. For two clusters A and B , it is defined as

$$J(A, B) = \frac{|A \cap B|}{|A \cup B|} = \frac{|A \cap B|}{|A| + |B| - |A \cap B|},$$

where $|A \cap B|$ is the number of observations shared by the two clusters and $|A \cup B|$ is the total number of observations assigned to at least one of them. The Jaccard index ranges from 0 to 1. A value of 0 indicates that the clusters contain no observations in common, a value of 1 indicates that their memberships are identical and intermediate values represent partial overlap.

Accuracy

Accuracy measures the proportion of correctly classified samples among all predictions. It is defined as:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN},$$

where TP and TN denote the number of true positive and true negative pre-

dictions, respectively, while FP and FN represent false positive and false negative predictions.

F1-score

The F1-score is the harmonic mean of precision and recall and provides a balance between these two metrics. Precision measures the proportion of correctly identified positive samples among all samples predicted as positive, while recall measures the proportion of correctly identified positive samples among all actual positive samples.

Precision and recall are defined as:

$$\text{Precision} = \frac{TP}{TP + FP},$$

$$\text{Recall} = \frac{TP}{TP + FN}.$$

The F1-score is then calculated as:

$$F1 = 2 \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} = \frac{2TP}{2TP + FP + FN}.$$

2.2 Graphs and Network Analysis

2.2.1 Graphs

A graph is a structure defined as a set of nodes (or vertices) connected by edges, where nodes represent the entities of interest and edges describe the relationships between them [24].

A graph can be formally represented as:

$$G = (V, E),$$

where V is the set of nodes and E is the set of edges connecting pairs of nodes.

Graphs can be either directed or undirected, depending on whether the relationship between two nodes has a specific direction. In an undirected graph, an edge between two nodes indicates a symmetric relationship, whereas in a directed graph, edges represent asymmetric relationships. Furthermore, edges can be either unweighted or weighted. In weighted graphs, each edge is associated with a numerical value representing the strength of the relationship between two nodes.

The structure of a graph can be represented through an adjacency matrix A , where each element A_{ij} describes the connection between nodes i and j . For an

unweighted graph, $A_{ij} = 1$ indicates the presence of an edge and $A_{ij} = 0$ indicates its absence, while in weighted graphs, A_{ij} contains the corresponding edge weight.

In neuroscience, graphs are often used, representing brain structures or regions as nodes and their structural or functional relationships as edges, enabling the study of brain organization as a network.

RBF Similarity

The Radial Basis Function (RBF) kernel [18] measures the similarity between two data points based on their distance in the feature space. It assigns high similarity to points that are close to each other and progressively lower similarity as their distance increases. The RBF similarity is defined as

$$K(\mathbf{x}, \mathbf{y}) = \exp \left(-\gamma \|\mathbf{x} - \mathbf{y}\|^2 \right),$$

where $\|\mathbf{x} - \mathbf{y}\|^2$ is the squared Euclidean distance between the two points and $\gamma > 0$ controls the rate at which similarity decreases with distance.

In this work, the RBF kernel is applied to the gray matter intensities to calculate the edge weights of connected nodes. Thus, for two connected nodes i and j , the edge weight was defined as

$$w_{ij} = \exp \left(-\gamma (x_i - x_j)^2 \right),$$

where x_i and x_j denote the gray matter intensities of nodes i and j , respectively. Consequently, nodes with similar gray matter intensities receive an edge weight close to 1, whereas larger differences in intensity result in weights closer to 0.

2.2.2 Multiplex Networks

A multiplex network is an extension of the classical graph representation that allows multiple types of relationships between the same set of nodes to be modeled simultaneously [25]. Instead of representing a system as a single graph, a multiplex network consists of multiple overlapped layers, where each layer describes a specific type of interaction.

Formally, a multiplex network can be represented as a set of graphs:

$$M = \{G^{(1)}, G^{(2)}, \dots, G^{(L)}\},$$

where L is the number of layers and each layer is defined as:

$$G^{(l)} = (V, E^{(l)}),$$

with V representing the common set of nodes and $E^{(l)}$ representing the edges in layer l . Thus, the same nodes are shared across layers, while the connections between them may differ depending on the type of relationship represented.

The adjacency matrix of a multiplex network can therefore be represented as a collection of layer-specific adjacency matrices:

$$A = \{A^{(1)}, A^{(2)}, \dots, A^{(L)}\},$$

where each matrix $A^{(l)}$ describes the connections between nodes within layer l . Notably there are no edges between layers.

2.2.3 Community Detection

Community detection is an unsupervised graph analysis technique aimed at identifying groups of nodes that are more densely connected to each other than to the rest of the network [26]. These groups, commonly referred to as communities, represent subsets of the network with a high degree of internal connectivity and relatively weaker connections with other groups.

Given a graph $G = (V, E)$, community detection algorithms aim to find a partition of the node set V into communities:

$$C = \{C_1, C_2, \dots, C_k\},$$

where each node belongs to one community and nodes within the same community have stronger relationships compared with nodes belonging to different communities.

Community detection can be done in single graphs as well as multiplex graphs [27].

Modularity and Louvain Algorithm

Louvain [28] is a greedy community-detection algorithm based on modularity optimization. Modularity measures the difference between the observed connectivity within communities and the connectivity expected under a null model with similar network properties but randomized connections [29]. It is defined as

$$Q(\gamma) = \frac{1}{2m} \sum_{ij} \left(A_{ij} - \gamma \frac{k_i k_j}{2m} \right) \delta(c_i, c_j),$$

where A_{ij} denotes the adjacency matrix, k_i and k_j are the degrees of nodes i and j , respectively, m is the total number of edges, and $\delta(c_i, c_j)$ equals 1 if nodes i and j belong to the same community and 0 otherwise. For an undirected weighted

network, the degree is replaced by the node strength, defined as $k_i = \sum_j A_{ij}$, while m corresponds to the total edge weight, $m = \frac{1}{2} \sum_{ij} A_{ij}$.

The resolution parameter γ controls the granularity of the detected communities: higher values favor smaller communities, whereas lower values favor larger communities.

A higher modularity value indicates a stronger community structure.

Louvain is one of the most widely used algorithms for detecting community structure in large networks due to its computational efficiency and scalability.

The algorithm follows a hierarchical optimization strategy composed of two main phases. In the first phase, each node is initially assigned to its own community. Nodes are then iteratively moved between neighboring communities, and each possible move is evaluated based on the resulting change in modularity. A node is assigned to the community that provides the largest modularity increase. This process is repeated until no further improvement in modularity can be achieved.

In the second phase, the identified communities are aggregated into super-nodes, creating a new coarse-grained network in which each node represents a community from the previous iteration. The modularity optimization process is then repeated on this reduced network. These two phases are iterated until modularity cannot be further increased, resulting in the final community partition.

Although the Louvain algorithm is highly efficient, it has some limitations, including the possibility of identifying poorly connected communities and sensitivity to the order in which nodes are processed. The Leiden algorithm [30] was introduced as an improvement over Louvain to overcome these limitations. It incorporates an additional refinement step between the two Louvain phases described above, ensuring that the resulting clusters are better connected and more stable. In practice, within each community found during the initial local moving phase the algorithm treats every node as its own separate group and repeat the optimization process. This time nodes are only allowed to merge with neighboring sub-communities that belong to the same larger community identified in the first phase. This prevents nodes from jumping across the network and creating disconnected pieces. The refined, well-connected sub-communities are then aggregated into super-nodes and used as the building blocks for the next network aggregation step.

Consensus Clustering

Due to the stochastic and greedy nature of the algorithm, the resulting community partitions are not fully deterministic, and different runs may produce slightly different solutions.

Consensus clustering [31] is a strategy used to improve the stability and robust-

ness of community assignments by combining the results obtained from multiple independent runs of the algorithm. The algorithm is repeated multiple times and/or varying resolution parameters, and the results are summarized into an $N \times N$ consensus matrix. Each element of this matrix represents the proportion of runs in which two nodes are assigned to the same community.

The consensus matrix is then used to construct a new weighted graph, where edge weights correspond to the consensus values between node pairs. Community detection is subsequently applied again to this new graph, and the process is iterated until convergence, defined as the point at which the community partition remains unchanged between consecutive iterations.

2.3 Feature Selection

Feature selection is the process of identifying and retaining the features that are most relevant to a machine learning task while removing irrelevant or redundant variables [18]. By reducing the dimensionality of the feature space, feature selection can improve model performance, reduce the risk of overfitting, and decrease computational requirements. This is particularly important in high-dimensional datasets where the number of features is large relative to the number of available samples.

Feature selection methods can generally be divided into three categories: *filter methods*, *wrapper methods*, and *embedded methods*.

1. **Filter methods** evaluate the relevance of individual features independently of the machine learning model. Features are selected according to statistical measures that quantify their association with the target variable. Common approaches include information gain, the χ^2 test, Fisher's score, Pearson's correlation coefficient, variance thresholding, and mean absolute difference. Filter methods are computationally inexpensive and can therefore be particularly useful for high-dimensional datasets. They are also independent of the choice of machine learning algorithm. However, because features are generally evaluated individually, these methods may fail to capture interactions between features that are relevant for prediction.
2. **Wrapper methods** evaluate subsets of features according to their effect on the performance of a specific machine learning model. A search strategy is used to iteratively add or remove features, and the subset producing the best model performance is selected. Common approaches include forward selection, backward elimination, and recursive feature elimination. Because wrapper methods directly account for the interaction between feature selection and

model performance, they can identify feature subsets that are well suited to a particular model. However, they are computationally expensive, particularly for high-dimensional datasets, and may increase the risk of overfitting.

3. **Embedded methods** perform feature selection as part of the model training process. The selection of relevant features is incorporated directly into the optimization procedure, allowing the model to identify informative features while being fitted. Examples include regularization-based methods, such as LASSO regression, and tree-based feature selection. Embedded methods provide a compromise between the computational efficiency of filter methods and the model-specific nature of wrapper methods. However, their applicability depends on the machine learning algorithm and the resulting feature selection may be more difficult to interpret independently of the model.

2.3.1 Univariate Feature Selection

Univariate feature selection is a filter-based approach in which each feature is evaluated independently according to its association with the target variable using a statistical scoring function. Features are then ranked by their scores, and only the highest-ranking features are retained. Selection may be based either on a fixed number of features, k , or on a predefined proportion of the original feature set.

In this study, mutual information was used as the scoring function. Mutual information quantifies the reduction in uncertainty about the class label obtained by observing a given feature [32]. Equivalently, it measures the statistical dependence between each feature and the class label, including potentially nonlinear relationships. Higher values indicate a stronger association with the target, whereas a value of zero indicates statistical independence.

2.4 Classification Algorithms

2.4.1 Support Vector Machine

Support Vector Machines (SVM) are supervised machine learning models used for classification. The main idea is to find a decision boundary that best separates data points belonging to different classes [33].

Formally, a Support Vector Machine learns a hyperplane defined as

$$\mathbf{w}^\top \mathbf{x} + b = 0,$$

which separates the classes while maximizing the margin, i.e., the distance between the hyperplane and the closest data points from each class.

In practice, data are often not perfectly linearly separable. For this reason, Support Vector Machines introduce a soft-margin formulation that allows classification errors through slack variables. A regularization parameter C controls the trade-off between maximizing the margin and minimizing classification errors.

The resulting optimization problem is given by:

$$\min_{\mathbf{w}, b, \xi} \frac{1}{2} \|\mathbf{w}\|^2 + C \sum_{i=1}^N \xi_i$$

subject to

$$y_i(\mathbf{w}^\top \mathbf{x}_i + b) \geq 1 - \xi_i, \quad \xi_i \geq 0, \quad i = 1, \dots, N.$$

where ξ_i is the slack variable associated with sample i

2.4.2 Logistic Regression

Logistic regression is another supervised machine learning algorithm for classification. Instead of predicting continuous outcomes, it models the log-odds of an event as a linear combination of one or more independent variables [34]. The resulting linear score is then passed through the sigmoid function to map it into a probability between 0 and 1, representing the likelihood that a given observation belongs to a specific class (e.g., 0 or 1, Yes or No).

Formally, the model is expressed as:

$$P(y = 1 \mid \mathbf{x}) = \sigma(\mathbf{w}^\top \mathbf{x} + b),$$

where $\sigma(\cdot)$ denotes the sigmoid function.

The model parameters are estimated by minimizing the cross-entropy loss function with an added regularization term to prevent overfitting:

$$\mathcal{L} = - \sum_{i=1}^N [y_i \log(p_i) + (1 - y_i) \log(1 - p_i)] + \lambda \Omega(\mathbf{w}),$$

where $\Omega(\mathbf{w})$ denotes the regularization term and λ controls its strength.

L1 regularization was chosen because it promotes sparsity in the learned weight vector by driving a subset of coefficients exactly to zero. This is particularly advantageous in high-dimensional settings like ours, as it performs implicit feature selection.

2.4.3 Random Forest and XGBoost

Decision Tree

A decision tree is a non-parametric supervised learning method that can be used for both classification and regression tasks [35]. Its objective is to learn a set of simple if-then decision rules from the input features in order to predict the value of a target variable. From a mathematical perspective, a decision tree can be interpreted as a piecewise constant approximation of the underlying function mapping inputs to outputs.

The model is constructed by recursively partitioning the feature space using axis-aligned splits. At each step, the algorithm selects the feature and threshold that most effectively reduce an impurity measure.

The tree structure emerges naturally from this recursive splitting procedure. The initial partition of the data is performed at the root node, which determines the first decision rule applied to all samples. Each subsequent split defines intermediate decision points that progressively refine the partition of the feature space into more homogeneous regions. These regions are represented by terminal nodes, or leaves, which provide the final prediction associated with each input sample.

Random Forest

Random Forests are bagging based models that combine multiple decision trees into an ensemble [36]. In this framework, N decision trees are trained independently on bootstrap samples of the data, and at each split a random subset of features is considered.

For classification task, the standard formulation aggregates the predictions via majority voting. However, the `scikit-learn` implementation we use instead combines trees by averaging their predicted class probabilities and selecting the class with the highest mean probability, rather than performing a hard vote over class labels.

In this work, node impurity of each decision tree is quantified using the Gini index, defined as:

$$G = 1 - \sum_{k=1}^K p_k^2,$$

where p_k represents the proportion of samples belonging to class k within a given node, and K is the total number of classes.

Gradient Boosting

Gradient Boosting is an ensemble learning technique that combines multiple weak learners, typically decision trees, to build a strong predictive model. Unlike bagging methods such as Random Forests, in which trees are trained independently, gradient boosting constructs trees sequentially. At each iteration, a new tree is trained to correct the errors made by the current ensemble, so that each successive model improves upon the previous one [37].

Given an input $\mathbf{x}$, the prediction of the ensemble after M iterations is

$$F_M(\mathbf{x}) = \sum_{m=1}^M \eta f_m(\mathbf{x}),$$

where $f_m(\mathbf{x})$ is the prediction of the m -th decision tree and η is the learning rate, which controls the contribution of each tree to the final model.

More generally, the model is built iteratively according to

$$F_m(\mathbf{x}) = F_{m-1}(\mathbf{x}) + \eta f_m(\mathbf{x}) \quad F_0(x) = 0,$$

where each new tree is fitted to the negative gradient of the loss function evaluated at the current predictions. For this reason, the algorithm is referred to as *gradient* boosting. By successively minimizing the loss function, the ensemble gradually improves its predictive performance.

XGBoost

Extreme Gradient Boosting (XGBoost) [38] is an optimized implementation of gradient boosting that improves both computational efficiency and predictive performance. At each iteration, XGBoost trains a new tree by minimizing the following regularized objective function:

$$\mathcal{L}^{(m)} = \sum_{i=1}^N l(y_i, \hat{y}_i^{(m-1)} + f_m(\mathbf{x}_i)) + \Omega(f_m),$$

where $l(\cdot)$ is a differentiable loss function measuring the prediction error, and

$$\Omega(f) = \gamma T + \frac{\lambda}{2} \|\mathbf{w}\|^2$$

is a regularization term that penalizes model complexity. Here, T is the number of leaves in the tree, $\mathbf{w}$ represents the leaf weights, and γ and λ are regularization parameters controlling tree complexity.

Rather than directly minimizing the objective, XGBoost employs a second-order Taylor approximation of the loss function, allowing both first- and second-order

derivatives to guide the optimization. This results in faster convergence and improved numerical stability compared with standard gradient boosting.

In addition to regularization, XGBoost incorporates several practical improvements, including shrinkage through a learning rate, row and column subsampling, efficient handling of missing values, and parallel tree construction. These features reduce overfitting while making the algorithm computationally efficient.

2.5 Explainable AI

Explainable AI is a field of research that explores methods that provide humans with the ability of intellectual oversight over AI algorithms. The main focus is on the reasoning behind the decisions or predictions made by the AI algorithms, to make them more understandable and transparent [39].

2.5.1 SHAP Values

The SHAP (SHapley Additive exPlanations) value [40] is one of the most widely used methods in explainable AI for interpreting machine learning models. It is based on the concept of the Shapley value [41], originally introduced by Lloyd Shapley in cooperative game theory as a principled method for fairly distributing the total gain or cost among a group of players according to their individual contributions.

In the context of machine learning, SHAP treats the input features as players in a cooperative game and quantifies the contribution of each feature to the model's prediction.

Specifically, SHAP explains an individual model output as the sum of a baseline value and feature-specific contributions. The baseline represents the expected model output under a reference, or background, distribution. Each SHAP value quantifies the contribution assigned to a feature toward the difference between the subject's model output and this baseline. Positive contributions increase the output associated with SBMA, whereas negative contributions decrease it. The SHAP value of a feature is computed as its average marginal contribution over all possible subsets of features. This provides a measure of the importance of each feature while accounting for interactions with the remaining features.

A key advantage of SHAP is that it satisfies the four desirable properties of Shapley values: *efficiency*, *symmetry*, *dummy player* (or *null player*), and *additivity* [41]. These axioms ensure that feature contributions are assigned in a mathematically consistent and fair manner.

Chapter 3

Materials and Methods

3.1 Dataset

The dataset consisted of 37 structural cortical MRI scans, including 20 patients with SBMA and 17 healthy controls in a Neuroimaging Informatics Technology Initiative (NIfTI) format.

High-resolution structural T1-weighted magnetic resonance imaging scans were acquired using a 3 T scanner equipped with a 32-channel head coil.

3.1.1 Preprocessing

MRI data were preprocessed using the Computational Anatomy Toolbox implemented in Statistical Parametric Mapping. The preprocessing pipeline included tissue segmentation to extract gray matter (GM) maps, spatial normalization to a common anatomical space, modulation to preserve regional tissue volumes after normalization, and spatial smoothing using an 8-mm full-width at half-maximum Gaussian kernel. These steps reduce inter-subject anatomical variability and improve the signal-to-noise ratio for subsequent analyses.

Following preprocessing, each subject was represented by a three-dimensional GM volume with dimensions $113 \times 137 \times 113$ voxels, where each voxel value corresponds to the local gray matter intensity.

For each subject, the gray matter images were parcellated into 1,000 regions (500 per hemisphere) according to the Schaefer cortical parcellation [42], with each region assigned to one of seven canonical functional brain networks [43]. For each parcel, the GM intensity was calculated as the mean intensity of all voxels belonging to that region, resulting in a feature vector representing the GM distribution across the 1,000 functional regions.

3.2 Community Detection Pipeline

3.2.1 Graph Construction

The graph-based analysis was implemented in Python. For each subject, an undirected weighted graph was constructed from the 1,000 parcel-wise GM intensity values obtained during preprocessing. Each node represented a parcel, and its spatial location was defined by the centroid of the voxels belonging to that parcel.

The graph topology was defined based on the spatial proximity of the nodes. An edge was established between two nodes when the Euclidean distance between their centroid coordinates was equal or below 15 spatial units. This threshold was selected to provide a sufficiently connected graph while maintaining computational feasibility. With this configuration, nodes had a mean of 26.97 edges (median: 27), with the number of edges per node ranging from 7 to 58. Edge weights were subsequently determined by the similarity between the regional gray matter (GM) values of connected nodes. Specifically, the similarity between two connected nodes was computed using an RBF kernel with $\gamma = 150$.

To determine an appropriate kernel width, the overall mean squared difference in GM values between parcels across all 37 patients was chosen to correspond to an RBF similarity weight of approximately 0.5. Solving

$$0.5 = \exp(-\gamma \cdot 0.005)$$

gives

$$\gamma = \frac{\ln(2)}{0.005} \approx 139.$$

Since the median squared edge difference (≈ 0.002) was substantially smaller than the mean, γ was increased slightly to 150. This choice makes the kernel more sensitive to the smaller intensity differences that characterize the majority of connected node pairs while preserving meaningful variation across the network.

3.2.2 Multiplex Construction

Following graph construction, two separate multiplex networks were generated: one representing the SBMA cohort and one representing the HC cohort. Each subject was represented as a separate layer within the corresponding multiplex network.

3.2.3 Community Detection

Community detection was subsequently performed using the Leiden algorithm implemented in the `leidenalg` Python package. For the multiplex networks, the `find_partition_multiplex` function was used with the `RBConfigurationVertexPartition` model and the previously defined edge weights.

Due to the stochastic nature of the Leiden algorithm, multiple runs may result in slightly different community assignments. To obtain stable community structures, the algorithm was repeated 30 times for each resolution parameter in the range 0.1 – 1.0 (step size 0.1). The resulting 30×10 partitions were combined into a consensus matrix, where each element represents the proportion of runs in which two nodes were assigned to the same community.

The consensus matrix was then thresholded at 0.7, retaining only node pairs that were assigned to the same community in at least 70% of the runs. The thresholded consensus matrix was subsequently used as the weighted adjacency matrix of a new graph. Community detection and consensus-matrix construction were repeated iteratively until convergence, defined as the point at which two consecutive iterations produced identical community assignments.

Since the consensus graph consisted of a single layer, the final community detection step was performed using the `find_partition` function rather than the multiplex version. The `RBConfigurationVertexPartition` model was again used, with edge weights derived from the consensus matrix. The algorithm was repeated across the same range of resolution parameters to account for stochastic variability.

After convergence, the identified communities were visualized by mapping the supervoxel assignments back onto the three-dimensional brain space. Representative SBMA and HC brain maps were generated, with each supervoxel colored according to its assigned community, allowing qualitative comparison of the spatial organization of brain communities between cohorts.

Finally, Jaccard similarity was used to quantify the overlap between communities identified in the SBMA and HC partitions. The resulting similarity matrix summarized the degree of correspondence between communities across groups. Communities showing little or no overlap with communities in the other group were subsequently examined to identify potential group-specific differences in brain network organization.

The overall pipeline can be visualized in Figure 3.1.

3.3 Classification Pipeline

3.3.1 Input Data

Two representations of the regional gray matter data were evaluated. The first consisted of the raw regional gray matter values, obtained by averaging the gray matter intensities of the voxels within each of the 1,000 atlas-defined regions, as described in the previous section. The second representation consisted of normalized regional GM values, in which the average GM value of each region was normalized by the subject's total gray matter. The latter was calculated as the sum of the GM values of all voxels in a patient MRI image and was used to account for differences in overall gray matter volume, for example due to age or other sources of inter-subject variability.

3.3.2 Feature Selection

To reduce the dimensionality of the feature space, the 10 most informative features were selected using the `SelectKBest` function from `scikit-learn`, with `mutual_info_classif` as the scoring function, and $k = 10$. This constitutes a univariate filter-based feature selection method, in which each feature is evaluated independently with respect to the target variable.

Mutual information is a non-negative measure of the statistical dependence between two random variables. It is equal to zero when the variables are statistically independent, while higher values indicate greater statistical dependence. The implementation in `scikit-learn` estimates mutual information using a non-parametric method based on entropy estimation from k -nearest-neighbour distances.

Feature selection was performed independently within each iteration of the LOOCV procedure. In each iteration, a standard scaler was fitted exclusively to the training data and then applied to both the training set and the held-out subject. Mutual information scores were calculated using only the scaled training data, and the 10 highest-scoring features were selected. The resulting feature subset was then applied to the held-out subject. This approach prevented information leakage by ensuring that the test subject did not influence either data normalization or feature selection.

3.3.3 Classification Algorithms

Four supervised learning algorithms were subsequently trained on both input representations within the same LOOCV framework: SVM, Logistic Regression, Random Forest, and XGBoost. The SVM, Logistic Regression, and Random Forest models were implemented using the `scikit-learn` Python package, while XGBoost was

implemented using the dedicated `xgboost` package.

The following model configurations were used:

- **Support Vector Machine:** A linear SVM was implemented using `sklearn.svm.SVC()` with linear kernel. Parameters not explicitly specified were kept at their default values.
- **Logistic Regression:** Logistic regression was implemented using `sklearn.linear_model.LogisticRegression()` with the 'liblinear' solver and an L1 penalty. The choice of L1 regularization was motivated by the high dimensionality of the feature space compared with the limited number of samples. Unlike L2 regularization, which continuously shrinks feature weights, L1 regularization encourages sparse solutions by driving some feature coefficients exactly to zero, effectively performing additional feature selection.
- **Random Forest:** A Random Forest classifier was implemented using `sklearn.ensemble.RandomForestClassifier()` with `random_state=4`. All remaining parameters were kept at their default values.
- **XGBoost:** An Extreme Gradient Boosting classifier was implemented using `xgboost.XGBClassifier()`. Parameters not explicitly specified were kept at their default values.

3.3.4 Interpretability Analysis

Feature contributions and SHAP values were quantified using the `shap` library. Explanations were generated within the leave-one-out cross-validation (LOOCV) procedure, such that preprocessing, feature selection, standardization, and model fitting were performed exclusively on the training folds. SHAP values were then computed for the corresponding held-out participant. Linear SVM and logistic-regression models were explained using `shap.LinearExplainer`, with the respective training fold used as the background dataset. Random-forest models were explained using `shap.TreeExplainer`, while XGBoost contributions were obtained using its package native TreeSHAP implementation. For binary classification, positive SHAP values represented contributions toward the SBMA class, whereas negative values represented contributions toward the healthy-control class. The explained output differed across classifiers: SVM contributions were expressed on the SVM margin scale, logistic-regression and XGBoost contributions on the log-odds scale, and random-forest contributions on the SBMA probability scale.

Because feature selection was repeated independently within each LOOCV fold, the resulting contributions were mapped back to the complete atlas feature space.

Thus, each fold yielded a contribution vector containing all atlas parcels, with a value of zero assigned to parcels that were not selected in that fold. This ensured that contributions could be consistently aggregated and compared across folds.

For global interpretation, four complementary measures were calculated:

- **Selection Frequency (SF):** the proportion of LOOCV folds in which a given region was selected as a feature included in the model.
- **Mean Absolute SHAP when Selected (MASS):** the mean absolute SHAP value of a region conditional on its selection. It was calculated as the sum of the absolute SHAP values across folds in which the region was selected, divided by the number of folds in which it was selected.
- **Mean Signed SHAP (MSS):** the mean signed SHAP value across participants,

$$\text{MSS}_j = \frac{1}{N} \sum_{i=1}^N \phi_{ij},$$

summarizing the average direction of a region's contribution. Values close to zero may, however, arise from cancellation between positive and negative subject-specific contributions.

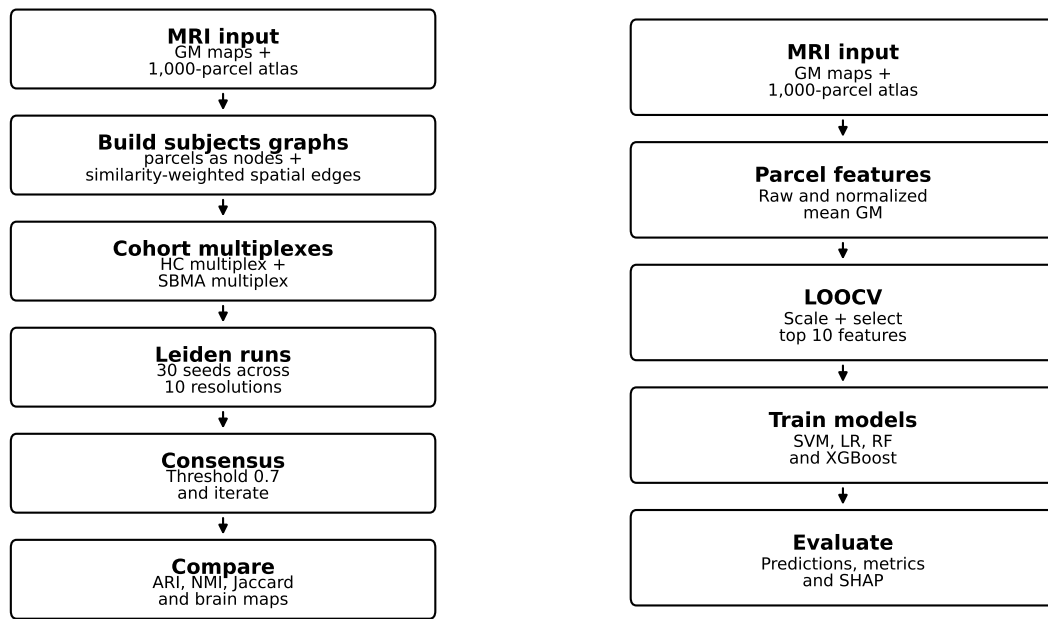
- **Mean Overall SHAP (MOS):** the mean absolute SHAP value across all participants,

$$\text{MOS}_j = \frac{1}{N} \sum_{i=1}^N |\phi_{ij}|,$$

providing a measure of the overall magnitude of a region's contribution across the dataset, including participants for whom the region was not selected.

Because the classifiers were explained on different output scales SHAP magnitudes were compared only within the same classifier and data representation.

The pipeline can be visualized in Figure 3.1b.



(a) Community-detection pipeline.

(b) Classification pipeline.

Figure 3.1: Overview of the two analytical pipelines.

Chapter 4

Results

4.1 Community Detection Results

The partitions identified by the Leiden algorithm differed between the two multiplex networks. Eight clusters were identified in the healthy control multiplex, whereas seven clusters were identified in the SBMA multiplex. Despite this difference in the number of communities, the overall organization of the two networks was highly similar. The Adjusted Rand Index and Normalized Mutual Information between the two partitions were 0.962 and 0.960, respectively, indicating a high degree of agreement between the clusterings. This similarity is expected, as the two cohorts represent the same underlying brain organization, with differences between groups expected to be localized to specific regions rather than affecting the overall organization of the brain. The resulting community partitions are visualized in Figure 4.1.

The correspondence between the two partitions is further illustrated by the Jaccard matrix in Figure 4.2.

The similarity matrix reveals an almost one-to-one correspondence between the HC and SBMA partitions. Most communities identified in the HC group overlap predominantly with a single SBMA community, with only a small number of parcels assigned differently. HC community 3, for example, corresponds exactly to SBMA community 3, with a Jaccard similarity of $J = 1.0$. Similarly, HC communities 0, 1, 2, 4, and 6 show near-complete correspondence with SBMA communities 0 ($J = 0.98$), 2 ($J = 0.98$), 1 ($J = 0.95$), 4 ($J = 0.99$), and 6 ($J = 0.96$), respectively. The correspondence between HC and SBMA community 5 is weaker ($J = 0.84$), because SBMA community 5 also includes 16 parcels from HC community 7 ($J = 0.12$).

The principal difference between the HC and SBMA partitions involved HC community 7 (Table 4.1). Of its 22 parcels, 16 were assigned to SBMA community 5 and the remaining six to SBMA community 1. Thus, HC community 7 did not

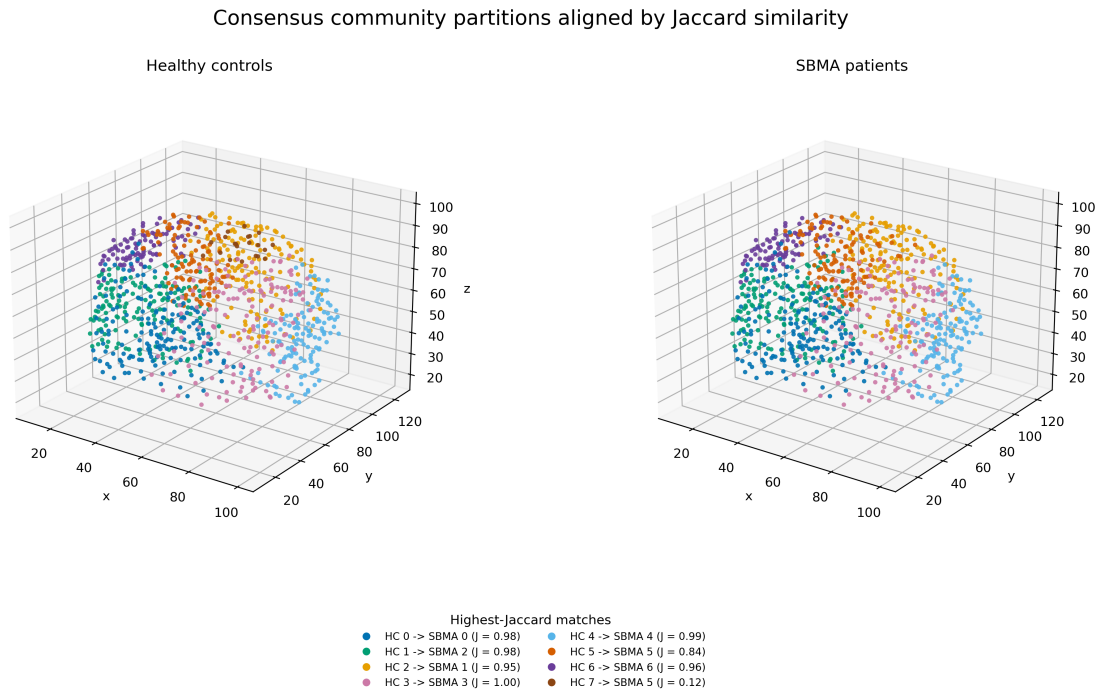


Figure 4.1: Consensus community partitions for healthy controls (left) and SBMA patients (right). Colours indicate highest Jaccard matches. HC cluster 7 is shown separately in brown because its match to SBMA cluster 5 is weak.

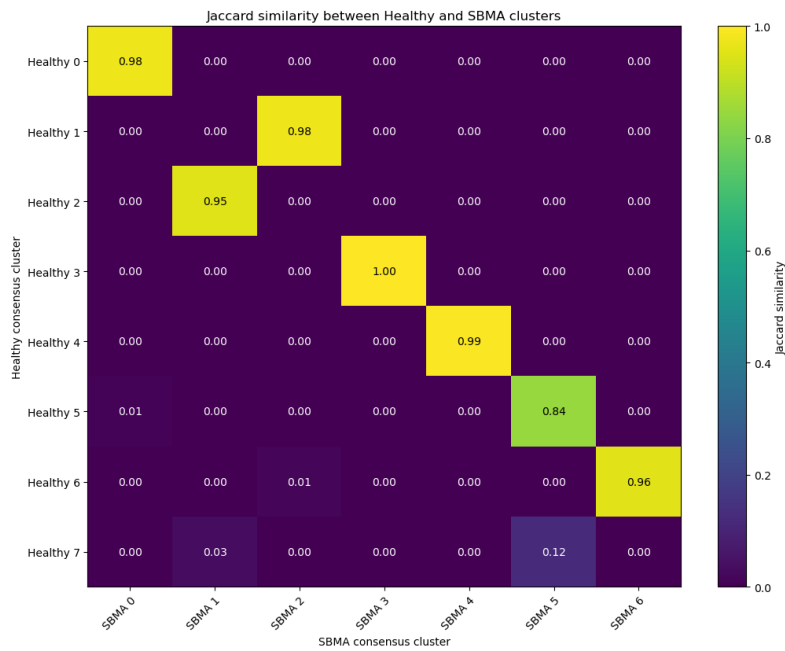


Figure 4.2: Jaccard similarity matrix

correspond to a single community in the SBMA partition but was redistributed across two larger SBMA communities.

Table 4.1: Parcels belonging to HC cluster 7.

Parcel name
7Networks_LH_SomMot_63
7Networks_LH_SomMot_64
7Networks_LH_SomMot_66
7Networks_LH_SomMot_67
7Networks_LH_SomMot_69
7Networks_LH_SomMot_70
7Networks_LH_SomMot_71
7Networks_LH_SomMot_72
7Networks_LH_SomMot_74
7Networks_LH_SomMot_75
7Networks_LH_SomMot_77
7Networks_LH_SomMot_79
7Networks_LH_SomMot_80
7Networks_LH_SomMot_81
7Networks_LH_SomMot_83
7Networks_LH_SomMot_84
7Networks_LH_SomMot_85
7Networks_LH_SomMot_89
7Networks_LH_SomMot_90
7Networks_LH_DorsAttn_Post_50
7Networks_LH_DorsAttn_FEF_4
7Networks_LH_DorsAttn_FEF_6

A closer examination HC community 7 indicates that it consists predominantly of parcels from the left somatosensory-motor network ($n = 19$), together with 2 parcels corresponding to the frontal eye field subdivision and one to the posterior subdivision of the left dorsal attention network. The redistribution of these parcels suggests that the principal differences between the two partitions are concentrated within these functional systems. In addition to this broader reorganization of HC community 7, only a small number of isolated parcels were assigned to different communities. Table 4.2 provides a parcel-level description of the observed reassignments. Six of the listed parcels are those belonging to HC community 7 that were assigned to SBMA cluster 1. The remaining seven parcels represent isolated reassignments outside HC community 7.

Together, the redistribution of community 7 and these isolated parcel-level reassignments account for the differences observed between the HC and SBMA community structures.

The full partitions are available in appendix A.

Table 4.2: Parcels whose SBMA cluster differs from the expected best-match cluster in HC.

Cluster name	HC cluster	SBMA cluster	Expected SBMA cluster
7Networks_LH_SomMot_63	7	1	5
7Networks_LH_SomMot_71	7	1	5
7Networks_LH_SomMot_75	7	1	5
7Networks_LH_SomMot_85	7	1	5
7Networks_LH_DorsAttn_FEF_4	7	1	5
7Networks_LH_DorsAttn_FEF_6	7	1	5
7Networks_LH_Default_pCunPCC_28	5	0	5
7Networks_RH_SomMot_76	5	0	5
7Networks_RH_DorsAttn_PrCv_2	6	2	6
7Networks_RH_Cont_PFCI_18	2	2	1
7Networks_RH_Cont_PFCI_20	6	2	6
7Networks_RH_Cont_PFCI_21	6	2	6
7Networks_RH_Default_pCunPCC_10	5	0	5

Overall, the findings indicate that the two partitions are highly similar and that the differences are primarily localized to the left somatosensory-motor network and, to a lesser extent, the dorsal attention network. Given the exploratory nature of the analysis, however, these results should be interpreted cautiously and cannot, on their own, establish a disease-specific alteration in brain community organization.

4.2 Classification Results

The classification performance obtained under the LOOCV framework is presented in Figure 4.3, with accuracy shown on the left and F1-score for the SBMA class on the right. The results are reported for the two input representations and the four classification algorithms.

Overall, the representation of the input data had a substantial effect on classification performance. Raw regional GM values resulted in the lowest performance across all classifiers, with accuracy ranging from 0.46 for the linear SVM to 0.57 for XGBoost. The corresponding F1-scores ranged from 0.47 to 0.60.

In contrast, normalized regional GM representations produced considerably higher classification performance. Logistic Regression and XGBoost achieved the highest performances, with accuracies of 0.84 and 0.81 and F1-scores of 0.84 and 0.84, respectively. Random Forest achieved a similar performance, with an accuracy of 0.78 and an F1-score of 0.82. Linear SVM obtained a lower accuracy of 0.76 and

an F1-score of 0.77.

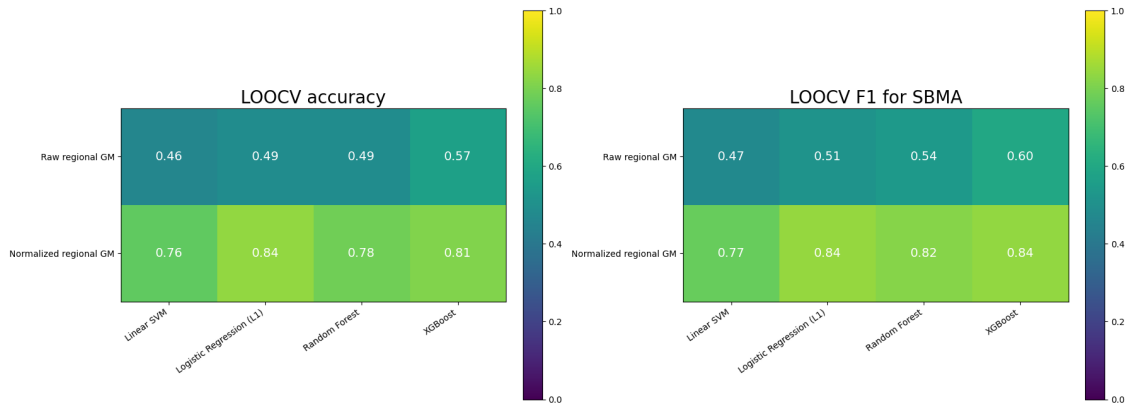


Figure 4.3: Models' accuracy and F1.

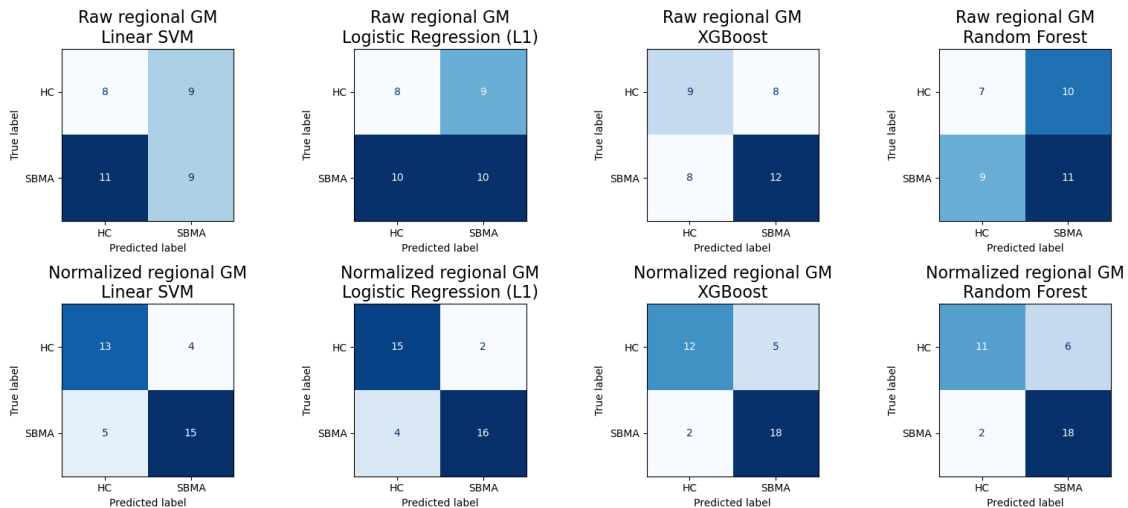


Figure 4.4: Models' Confusion Matrices.

Figure 4.4 presents the confusion matrices obtained for the 17 healthy controls and 20 patients with SBMA. XGBoost provided the highest and most balanced classification performance, correctly identifying 9 of 17 HC and 12 of 20 patients with SBMA (sensitivity: 60.0%; specificity: 52.9%). Random forest achieved the second-highest SBMA sensitivity, correctly classifying 11 of 20 patients (55.0%), but showed the lowest specificity, correctly identifying only 7 of 17 HC (41.2%). Logistic regression correctly classified 8 HC and 10 patients with SBMA (sensitivity: 50.0%; specificity: 47.1%), while linear SVM showed the lowest sensitivity, correctly identifying 9 of 20 patients with SBMA (45.0%).

Overall classification performance improved substantially when normalized regional GM measures were considered. L1-regularized logistic regression achieved the highest and most balanced performance, correctly identifying 15 of 17 HC and

16 of 20 patients with SBMA (sensitivity: 80.0%; specificity: 88.2%). Linear SVM also showed relatively balanced performance, correctly classifying 13 HC and 15 patients with SBMA (sensitivity: 75.0%; specificity: 76.5%). XGBoost and random forest achieved the highest SBMA sensitivity, each correctly identifying 18 of 20 patients (90.0%). However, XGBoost had higher specificity than random forest (70.6% versus 64.7%), misclassifying 5 rather than 6 HC as having SBMA. Thus, L1-regularized logistic regression provided the best overall balance between sensitivity and specificity, whereas XGBoost may be preferable when minimizing false negatives is the primary objective.

Figure 4.5 and Tables 4.3, 4.4 present the 15 features most frequently ranked among the 10 features with the highest mutual information across the LOOCV iterations, together with their respective selection frequencies per input data. Because this filter-based feature-selection method was applied before and independently of model fitting, the same set of candidate features was used for all classification models. The features selected from the raw and normalized regional GM measures differed substantially. This finding indicates that normalization alters the information captured by individual regions. Raw regional GM measures may reflect both region-specific differences and global variation in brain size or total GM volume, whereas normalized measures emphasize the relative distribution of GM across regions. Consequently, the features selected from normalized data may identify regional alterations that are less influenced by global atrophy.

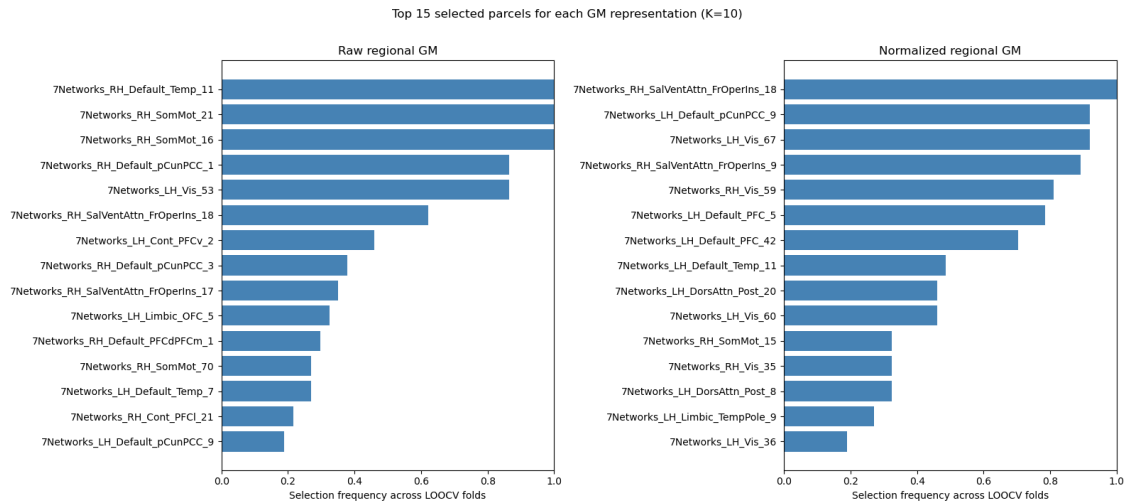


Figure 4.5: Most frequently selected features for GM representation.

Given the superior classification performance achieved using normalized regional GM measures, subsequent analyses focus exclusively on results obtained from these features.

SHAP analysis was performed at both global and subject-specific levels. In

Table 4.3: Most frequently selected features - Raw regional GM.

Rank	Parcel name	Times selected	Frequency (%)
1	7Networks_RH_SomMot_16	37	100.0
2	7Networks_RH_SomMot_21	37	100.0
3	7Networks_RH_Default_Temp_11	37	100.0
4	7Networks_LH_Vis_53	32	86.5
5	7Networks_RH_Default_pCunPCC_1	32	86.5
6	7Networks_RH_SalVentAttn_FrOperIns_18	23	62.2
7	7Networks_LH_Cont_PFCv_2	17	45.9
8	7Networks_RH_Default_pCunPCC_3	14	37.8
9	7Networks_RH_SalVentAttn_FrOperIns_17	13	35.1
10	7Networks_LH_Limbic_OFC_5	12	32.4
11	7Networks_RH_Default_PFCdPFCm_1	11	29.7
12	7Networks_LH_Default_Temp_7	10	27.0
13	7Networks_RH_SomMot_70	10	27.0
14	7Networks_RH_Cont_PFCI_21	8	21.6
15	7Networks_LH_Default_pCunPCC_9	7	18.9

Table 4.4: Most frequently selected features - normalized regional GM.

Rank	Parcel name	Times selected	Frequency (%)
1	7Networks_RH_SalVentAttn_FrOperIns_18	37	100.0
2	7Networks_LH_Vis_67	34	91.9
3	7Networks_LH_Default_pCunPCC_9	34	91.9
4	7Networks_RH_SalVentAttn_FrOperIns_9	33	89.2
5	7Networks_RH_Vis_59	30	81.1
6	7Networks_LH_Default_PFC_5	29	78.4
7	7Networks_LH_Default_PFC_42	26	70.3
8	7Networks_LH_Default_Temp_11	18	48.6
9	7Networks_LH_Vis_60	17	45.9
10	7Networks_LH_DorsAttn_Post_20	17	45.9
11	7Networks_LH_DorsAttn_Post_8	12	32.4
12	7Networks_RH_Vis_35	12	32.4
13	7Networks_RH_SomMot_15	12	32.4
14	7Networks_LH_Limbic_TempPole_9	10	27.0
15	7Networks_LH_Vis_36	7	18.9
16	7Networks_RH_SalVentAttn_FrOperIns_14	7	18.9
17	7Networks_RH_SalVentAttn_FrOperIns_19	5	13.5
18	7Networks_RH_Cont_Temp_2	5	13.5
19	7Networks_RH_Cont_PFCv_1	5	13.5
20	7Networks_RH_Vis_21	4	10.8

general, explanatory analyses are often performed by retraining the final selected model on the complete dataset after model selection. However, this approach may provide optimistic interpretations because the same subjects are used both for model fitting and explanation, therefore it was decided to perform SHAP analysis within the LOOCV framework instead, ensuring robustness of the interpretation.

Table 4.5 presents the feature-wise analyses for each classifier. For each model, only the 15 most important features, ranked according to their MOS value, are reported. The tables also provide the MASS, and MSS for each feature. For MSS, positive values indicate that the feature contributes, on average, toward the SBMA class, whereas negative values indicate a contribution toward the healthy-control class.

SHAP magnitudes should only be compared within the same model because model outputs use different scales (SVM margin, logistic/XGBoost log-odds, and Random Forest probability).

MOS, MASS, and MSS can be visualized in Figure 4.6, 4.7, and 4.8 respectively.

Table 4.5: Top 15 features for each classification model.

Rank	Feature	MOS	MASS	MSS
SVM top 15 features				
1	7Networks_RH_SalVentAttn_FrOperIns_18	1.174	1.174	−0.021
2	7Networks_RH_SalVentAttn_FrOperIns_9	0.651	0.730	0.142
3	7Networks_LH_Default_PFC_42	0.429	0.610	0.084
4	7Networks_LH_Default_pCunPCC_9	0.364	0.396	0.013
5	7Networks_LH_Default_Temp_11	0.334	0.687	0.039
6	7Networks_LH_Vis_67	0.312	0.340	−0.044
7	7Networks_LH_Default_PFC_5	0.303	0.387	0.096
8	7Networks_RH_Vis_59	0.218	0.269	−0.002
9	7Networks_LH_Vis_60	0.144	0.314	−0.031
10	7Networks_LH_Vis_36	0.080	0.423	−0.056
11	7Networks_RH_SomMot_15	0.075	0.232	−0.016
12	7Networks_LH_DorsAttn_Post_20	0.072	0.158	−0.001
13	7Networks_LH_DorsAttn_Post_8	0.063	0.196	−0.019
14	7Networks_LH_SalVentAttn_Med_9	0.060	1.104	−0.060
15	7Networks_LH_Limbic_TempPole_9	0.058	0.215	−0.026
Logistic Regression top 15 features				
1	7Networks_RH_SalVentAttn_FrOperIns_18	1.449	1.449	−0.097
2	7Networks_RH_SalVentAttn_FrOperIns_9	0.688	0.771	0.145
3	7Networks_LH_Default_PFC_42	0.546	0.777	0.130
4	7Networks_LH_Default_pCunPCC_9	0.364	0.396	−0.008

Rank	Feature	MOS	MASS	MSS
5	7Networks_LH_Vis_67	0.349	0.379	-0.055
6	7Networks_LH_Default_Temp_11	0.285	0.585	0.028
7	7Networks_LH_Default_PFC_5	0.255	0.326	0.029
8	7Networks_RH_Vis_59	0.184	0.227	0.000
9	7Networks_RH_SalVentAttn_FrOperIns_19	0.068	0.505	0.015
10	7Networks_LH_Vis_36	0.063	0.330	-0.045
11	7Networks_LH_SalVentAttn_Med_9	0.058	1.080	-0.058
12	7Networks_LH_DorsAttn_Post_8	0.042	0.129	-0.009
13	7Networks_RH_Vis_21	0.027	0.248	0.003
14	7Networks_RH_Cont_Temp_2	0.015	0.114	0.002
15	7Networks_LH_Limbic_TempPole_9	0.015	0.056	-0.015

Random Forest top 15 features

1	7Networks_RH_SalVentAttn_FrOperIns_18	0.119	0.119	0.003
2	7Networks_RH_SalVentAttn_FrOperIns_9	0.091	0.102	0.025
3	7Networks_LH_Default_PFC_42	0.044	0.063	0.011
4	7Networks_LH_Default_PFC_5	0.036	0.045	0.009
5	7Networks_RH_Vis_59	0.034	0.042	0.002
6	7Networks_LH_Default_pCunPCC_9	0.026	0.028	0.001
7	7Networks_LH_Vis_67	0.020	0.022	0.001
8	7Networks_RH_SomMot_15	0.017	0.052	0.006
9	7Networks_LH_Default_Temp_11	0.015	0.030	0.000
10	7Networks_LH_DorsAttn_Post_8	0.014	0.044	0.001
11	7Networks_RH_SalVentAttn_FrOperIns_14	0.010	0.054	0.005
12	7Networks_RH_SalVentAttn_FrOperIns_19	0.010	0.072	0.001
13	7Networks_LH_DorsAttn_Post_20	0.009	0.020	0.001
14	7Networks_RH_Vis_35	0.007	0.022	-0.003
15	7Networks_RH_Cont_Temp_2	0.006	0.045	0.003

XGBoost top 15 features

1	7Networks_RH_SalVentAttn_FrOperIns_18	0.945	0.945	0.057
2	7Networks_RH_SalVentAttn_FrOperIns_9	0.811	0.909	0.184
3	7Networks_LH_Default_PFC_42	0.404	0.575	0.196
4	7Networks_LH_Default_pCunPCC_9	0.150	0.163	-0.010
5	7Networks_LH_Default_Temp_11	0.135	0.278	-0.021
6	7Networks_LH_Vis_67	0.133	0.144	0.017
7	7Networks_LH_DorsAttn_Post_8	0.112	0.344	-0.025
8	7Networks_LH_Default_PFC_5	0.107	0.137	-0.015
9	7Networks_RH_Vis_59	0.107	0.131	-0.010
10	7Networks_RH_SomMot_15	0.102	0.313	0.054
11	7Networks_RH_Vis_35	0.041	0.127	-0.023

Rank	Feature	MOS	MASS	MSS
12	7Networks_LH_SomMot_80	0.037	0.693	0.037
13	7Networks_LH_DorsAttn_Post_20	0.037	0.081	-0.018
14	7Networks_RH_Cont_Temp_2	0.027	0.198	0.006
15	7Networks_RH_SalVentAttn_FrOperIns_14	0.024	0.126	0.005

Abbreviations: MOS: mean overall SHAP; MASS: mean absolute SHAP when selected; MSS: mean signed SHAP.

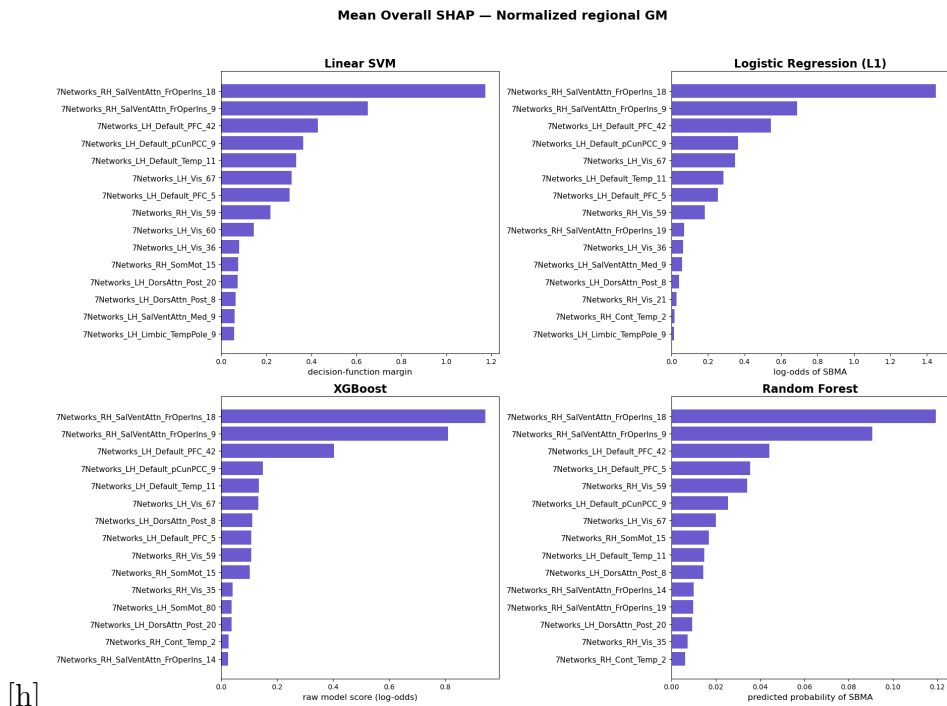


Figure 4.6: Mean Overall SHAP for each model.

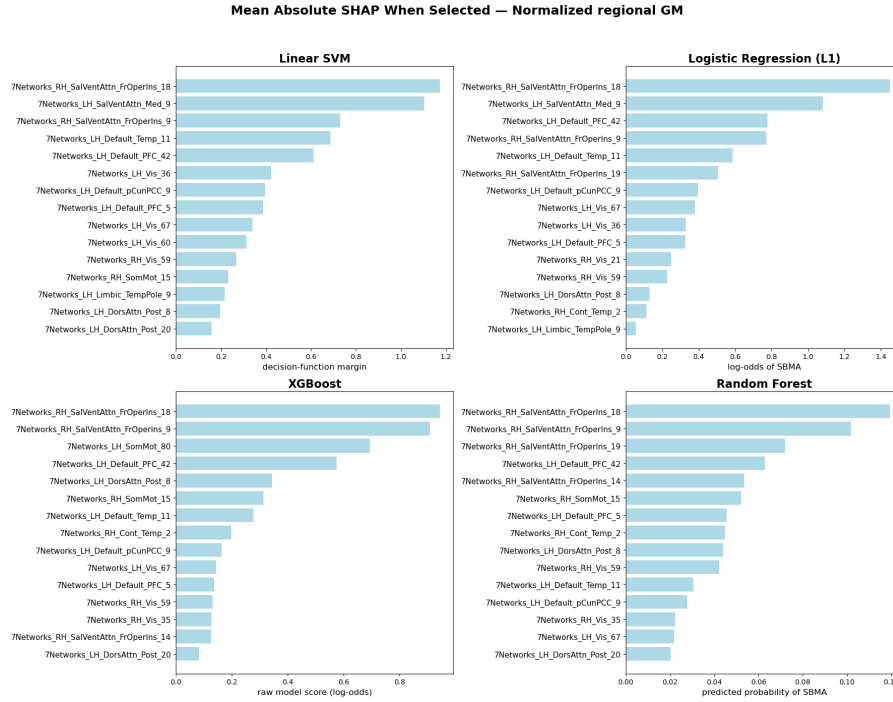


Figure 4.7: Mean Absolute SHAP when selected for each model.

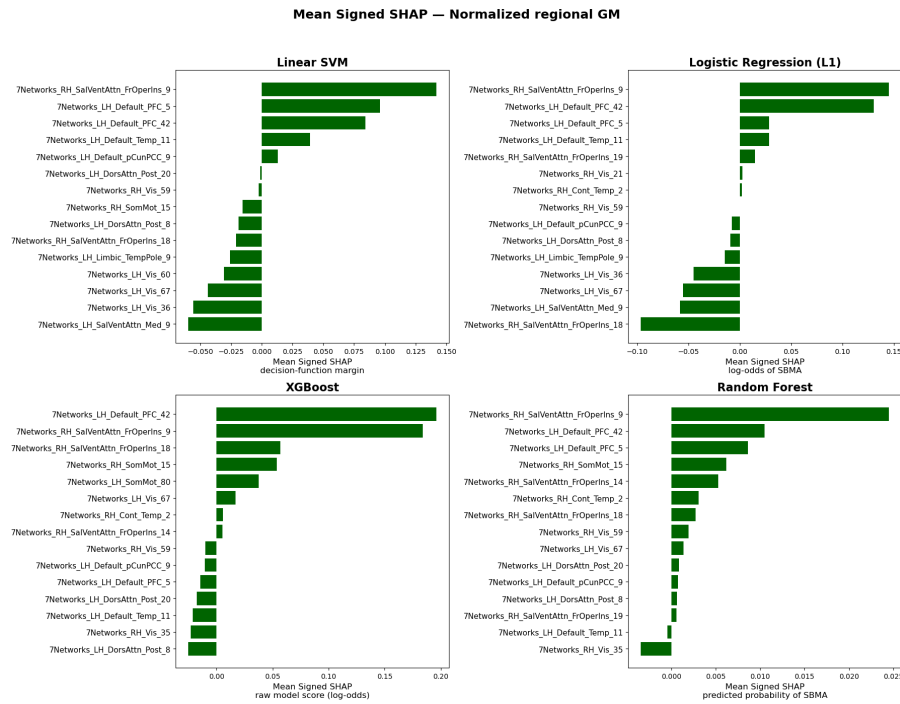


Figure 4.8: Mean Signed SHAP for each model.

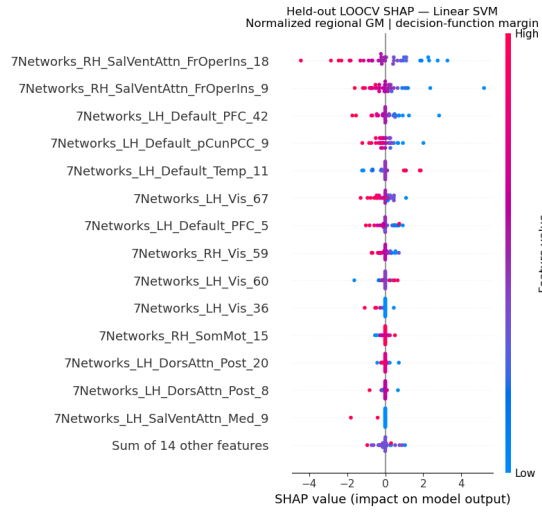
Larger absolute SHAP values indicate a greater influence of a component on the model output, irrespective of the direction of its contribution. However, a high MASS value should be interpreted in conjunction with the selection frequency. A component may exhibit a large contribution when selected but be selected in only a small proportion of LOOCV folds, suggesting that its importance is not stable

and may therefore be less informative at the global level.

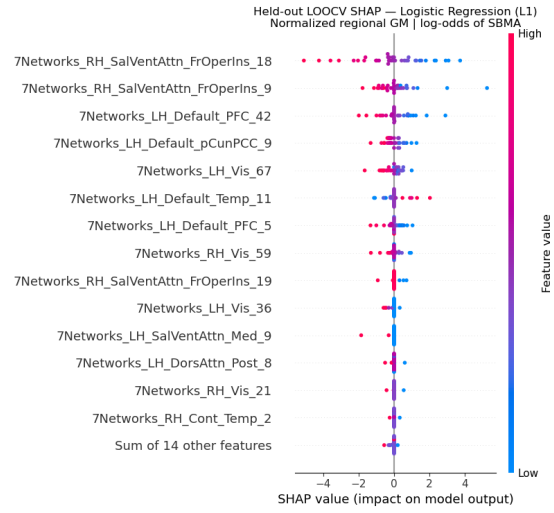
Beeswarm plots provide a simultaneous representation of the magnitude and direction of feature contributions, while indirectly accounting for selection frequency across LOOCV iterations. Positive SHAP values indicate that a feature shifts the prediction towards the SBMA class, whereas negative values shift it towards the healthy control class. Each point represents one held-out subject, and its colour indicates the corresponding standardized feature value, with red denoting higher values and blue denoting lower values. Features are ordered by their mean overall absolute SHAP value (MOS), with the most influential features displayed at the top. Because features not selected in a given LOOCV fold were assigned a SHAP value of zero, MOS reflects both the strength of a feature's contribution and how consistently it was selected across folds.

As shown in Figures 4.9a, 4.9b, 4.9c, and 4.9d, all four models identified `7Networks_RH_SalVentAttn_FrOperIns_18` as the most influential feature, followed by `7Networks_RH_SalVentAttn_FrOperIns_9` and `7Networks_LH_Default_PFC_42`. For these features, lower normalized GM values generally produced positive SHAP values, shifting predictions towards SBMA, whereas higher values produced negative SHAP values, shifting predictions towards the healthy control class. A similar, although weaker, pattern was consistently observed across classifiers for `7Networks_LH_Default_pCunPCC_9`, `7Networks_LH_Vis_67`, `7Networks_LH_Default_PFC_5`, and `7Networks_RH_Vis_59`. In contrast, `7Networks_LH_Default_Temp_11` consistently showed the opposite pattern: higher normalized GM values shifted predictions towards SBMA, whereas lower values shifted them towards the healthy control class. The remaining features either exhibited less consistent patterns, potentially reflecting nonlinear or subject-specific effects, or had SHAP values concentrated around zero, indicating relatively small contributions to the model predictions.

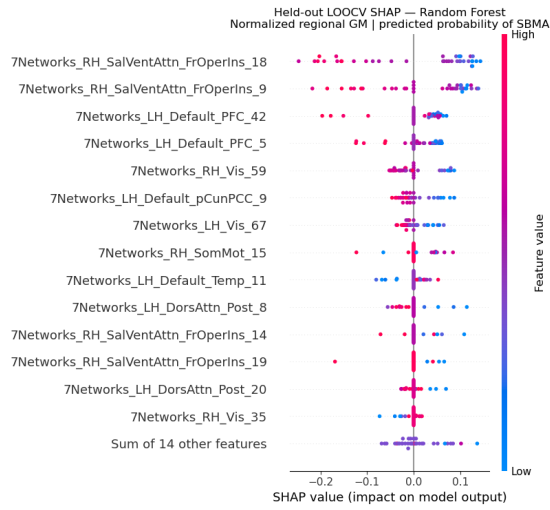
Overall, the SHAP analyses suggest that SBMA classification is predominantly associated with lower normalized GM values. This association was particularly evident in the frontal operculum–insula parcels of the salience/ventral-attention network and in prefrontal regions of the default-mode network. Conversely, the left temporal component of the default-mode network showed higher normalized GM values compared to HC subjects. These findings should be interpreted as model-derived associations rather than direct evidence of atrophy.



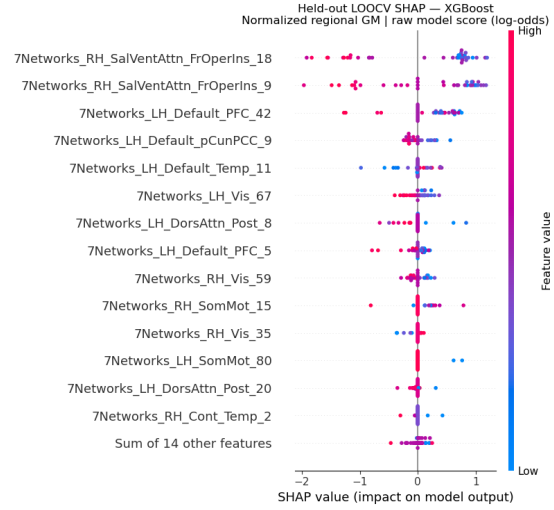
(a) Linear SVM.



(b) Logistic regression.



(c) Random forest.



(d) XGBoost.

Figure 4.9: Beeswarm plots showing the SHAP values obtained from the held-out LOOCV predictions using normalized regional GM measures.

Chapter 5

Discussions and Conclusions

Spinal and Bulbar Muscular Atrophy is a rare neurodegenerative disease primarily affecting the motor system, with well-established involvement of the spinal cord and associated white matter tracts. Due to its genetic basis, research has predominantly focused on the molecular and biological mechanisms underlying the disease. In contrast, relatively few studies have investigated potential brain involvement, and existing findings remain limited and inconclusive. To our knowledge, this is the first study to address this question using machine-learning methods.

5.1 Methodological Recap

Two complementary analytical pipelines were developed: a community detection pipeline and a classification pipeline.

In the community detection analysis, T1-weighted MRI data were used to construct a graph for each subject, with nodes representing brain parcels from the Schaefer cortical parcellation and edge weights representing the similarity in gray matter intensity between spatially connected parcels. Separate multiplex networks were constructed for the healthy control and SBMA groups, and community detection was performed to investigate whether the two groups exhibited different patterns of brain organization. Overall, the two partitions showed a high degree of correspondence, with differences primarily concentrated in a small set of regions within the left somatomotor cortex and, to a lesser extent, the dorsal attention network.

In the classification pipeline, the dimensionality of the regional MRI data was reduced using a univariate filter method based on mutual information, retaining the 10 regions most strongly associated with the class labels. Four supervised machine learning classifiers were then evaluated to distinguish SBMA patients from healthy controls. After demonstrating that the models could discriminate between

the two groups, SHAP analysis was used to investigate the features contributing most strongly to the predictions. The results indicated that lower proportional gray matter in parcels 18 and 9 of the frontal operculum–insula region of the salience/ventral attention network, as well as in parcel 42 of the default mode network in the prefrontal cortex, was associated with the SBMA class. To a lesser extent, higher proportional gray matter in parcel 11 of the left temporal component of the default mode network was also associated with SBMA.

5.2 Interpretation

At first glance, the results may appear to provide conflicting answers regarding where gray matter differences between patients with SBMA and healthy controls are. However, this apparent discrepancy reflects methodological differences between the analytical pipelines and the distinct properties assessed by each approach, rather than contradictory evidence.

From a methodological perspective, classification is a supervised, subject-level approach that identifies normalized regional GM values that help distinguish patients with SBMA from healthy controls. Multiplex community detection, by contrast, is a parcel-level approach that evaluates patterns of GM similarity among spatially connected regions. A parcel may therefore provide strong discriminatory information without changing its community membership. Conversely, a parcel showing only a modest difference in GM intensity between groups may be assigned to a different community if its relationship with neighboring parcels changes. Community-detection findings should consequently be interpreted as evidence of altered interregional GM organization, rather than as direct evidence of differences in absolute GM intensity.

Within the community detection analysis, the high adjusted Rand index and normalized mutual information values indicated that the overall community structures of the SBMA and HC groups were largely similar. This broad similarity was expected, given that the fundamental anatomical organization of the brain is shared between the two groups. Differences were limited to a relatively small proportion of parcels, primarily within the somatosensory–motor network and, to a lesser extent, the frontal eye field of the dorsal attention network. The anatomical distribution of these differences is biologically plausible in the context of a disease primarily affecting the motor system.

A complementary biological explanation is that SBMA may affect different parcels within the somatosensory–motor network across individual patients. Such spatial heterogeneity could reduce the consistency of parcel-level abnormalities and

limit classification performance.

In contrast, the smaller subset of parcels identified by the original classification pipeline exhibited more consistent discriminative patterns following normalization for total GM volume. These parcels may capture localized manifestations of disease-related neuroanatomical alterations that are obscured when gray-matter values are averaged across an entire network. However, the reliability and generalizability of these findings require validation in larger, independent datasets.

Overall, these findings demonstrate the potential of machine learning approaches for investigating brain involvement in SBMA and highlight several regions of interest that warrant further investigation at the neurological level.

5.3 Limitation and Future Research

The present study has several important limitations. The most general limitation is the small sample size, which limits the robustness and generalizability of the findings, particularly for the classification task. With a limited number of subjects, individual observations can have a substantial influence on model performance, resulting in higher variability and less stable estimates of predictive performance. This issue is particularly relevant in the present study given the high dimensionality of the MRI data relative to the number of available subjects.

For the classification pipeline, an additional limitation is the use of univariate mutual information for feature selection. Because each feature is evaluated independently with respect to the target variable, interactions between features are not considered. Consequently, correlated or redundant features may be selected, while combinations of features that are informative only through their interactions may be overlooked.

For the community detection pipeline, the construction of the graphs was strongly influenced by the choice of methodological parameters. In particular, the use of a fixed spatial distance threshold can introduce a bias related to the size and spatial distribution of the brain parcels. Larger parcels may have centroids that are farther from neighbouring centroids, resulting in fewer connections, whereas smaller adjacent parcels may have a larger number of neighbours. Furthermore, edge weights were defined using an RBF similarity between regional gray matter values, making the resulting edges' strength dependent on the choice of the γ parameter. These methodological choices may therefore influence the communities detected by the Leiden algorithm.

Future studies could address these limitations by using more directly observable measures to construct the brain networks. For example, functional MRI data could

be used to define edges according to the functional relationship between brain regions, such as their temporal correlation or co-activation. Edge weights could then reflect the strength of functional connectivity rather than a similarity measure based on structural gray matter intensity. This could provide a less parameter-dependent representation of brain organization.

Another possible improvement concerns the correction for age-related variation in gray matter. Rather than relying only on proportional normalization, future studies could explicitly model the expected gray matter volume of each brain region as a function of age. A linear regression model, for example, could be fitted using a large independent dataset of healthy individuals to estimate the expected gray matter volume for each region at a given age. The difference between the observed and expected gray matter volume could then be used as a residual measure of regional structural abnormality. These residuals could subsequently be used as input for the classification analysis, potentially allowing disease-related alterations to be distinguished more directly from normal age-related variation.

Overall, the findings suggest the presence of differences in regional GM patterns and interregional GM organization between SBMA and HC MRI scans. The classification and community-detection analyses provide complementary perspectives on these differences, although further investigation is required to confirm their biological and clinical significance.

Appendix A

Community Partitions

A.1 Community Partitions for HC

Table A.1: Parcel families and parcel numbers in each Healthy Control cluster.

Parcel family	Parcel numbers	Cluster
7Networks_LH_Cont_pCun	1, 2, 3	0
7Networks_LH_Default_PHC	3	0
7Networks_LH_Default_pCunPCC	1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 12, 13, 15, 18, 24, 30	0
7Networks_LH_Vis	4, 5, 8, 10, 11, 13, 14, 16, 17, 19, 20, 22, 23, 25, 26, 27, 31, 32, 35, 37, 38, 39, 41, 43, 46, 47, 48, 50, 51, 52, 55, 56, 57, 60, 62, 63, 66, 67, 69, 71, 73, 75	0
7Networks_RH_Cont_Par	4, 13	0
7Networks_RH_Cont_pCun	1, 3, 5	0
7Networks_RH_Default_Par	16	0
7Networks_RH_Default_pCunPCC	1, 2, 3, 4, 6, 7, 9, 15	0
7Networks_RH_DorsAttn_Post	2, 12, 13, 15, 16, 19, 21, 22, 24, 27, 32, 37, 39, 40, 43	0

Continued on next page

Table A.1 – continued

Parcel family	Parcel numbers	Cluster
7Networks_RH_Vis	1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 22, 23, 24, 25, 27, 28, 29, 30, 31, 32, 33, 35, 36, 37, 38, 40, 41, 42, 43, 44, 45, 46, 47, 48, 50, 51, 52, 53, 54, 55, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81	0
7Networks_RH_Cont_PFC1	7, 10, 12, 13, 15, 16	1
7Networks_RH_Cont_PFCv	1, 2	1
7Networks_RH_Cont_Par	1	1
7Networks_RH_Cont_Temp	1, 2, 3, 4	1
7Networks_RH_Default_PFCv	1, 2, 3, 4, 5, 6, 7, 8, 9, 10	1
7Networks_RH_Default_Par	1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 13	1
7Networks_RH_Default_Temp	1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18	1
7Networks_RH_DorsAttn_Post	1, 3, 4, 5, 6, 7, 8, 9, 10, 11	1
7Networks_RH_DorsAttn_PrCv	1	1
7Networks_RH_Limbic_OFC	1, 6	1
7Networks_RH_Limbic_TempPole	1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 17	1
7Networks_RH_SalVentAttn_FrOperI ns	1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24	1
7Networks_RH_SalVentAttn_TempOcc Par	1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 13, 14	1

Continued on next page

Table A.1 – continued

Parcel family	Parcel numbers	Cluster
7Networks_RH_SomMot	1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34	1
7Networks_RH_Vis	21, 26, 34, 39, 49, 56	1
7Networks_LH_Cont_Cing	1, 4, 6	2
7Networks_LH_Cont_OFC	1	2
7Networks_LH_Cont_PFCd	1	2
7Networks_LH_Cont_PFCl	5, 12, 19	2
7Networks_LH_Cont_PFCmp	1, 2	2
7Networks_LH_Default_PFC	5, 7, 9, 10, 12, 13, 14, 16, 19, 20, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 37, 38, 39, 40, 41, 43, 44, 45, 46, 47, 48, 49, 50	2
7Networks_LH_DorsAttn_FEF	2, 5, 7	2
7Networks_LH_Limbic_OFC	2, 3, 4, 6, 8, 9, 10, 11, 12, 13, 14	2
7Networks_LH_SalVentAttn_Med	1, 2, 3, 4, 6, 10, 15, 17, 18, 19	2
7Networks_LH_SalVentAttn_PFCl	1, 2	2
7Networks_LH_SomMot	42, 47, 55, 65, 82	2
7Networks_RH_Cont_Cing	2	2
7Networks_RH_Cont_PFCl	1, 2, 3, 4, 5, 6, 8, 9, 11, 14, 17, 18, 19, 22, 24, 28, 32, 33, 34	2
7Networks_RH_Cont_PFCmp	1, 2, 3, 4, 5	2
7Networks_RH_Default_PFCdPFCm	1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24	2
7Networks_RH_Limbic_OFC	2, 3, 4, 5, 7, 8, 9, 10, 11, 12, 13, 14	2
7Networks_RH_SalVentAttn_Med	1, 2, 3, 4, 5, 11, 15, 17, 19	2

Continued on next page

Table A.1 – continued

Parcel family	Parcel numbers	Cluster
7Networks_RH_SalVentAttn_PFC1	1, 2	2
7Networks_RH_SomMot	45, 52, 57, 64, 83, 91	2
7Networks_LH_Cont_Par	1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15	3
7Networks_LH_Cont_Temp	2, 3, 4	3
7Networks_LH_Default_PHC	1	3
7Networks_LH_Default_Par	1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19	3
7Networks_LH_Default_Temp	20, 22	3
7Networks_LH_DorsAttn_FEF	3	3
7Networks_LH_DorsAttn_Post	1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 42, 43, 46, 47, 48	3
7Networks_LH_SalVentAttn_ParOper	1, 2, 4, 6, 8, 9	3
7Networks_LH_SalVentAttn_TempOcc	1, 2	3
7Networks_LH_SomMot	19, 23, 39, 40, 41, 44, 45, 46, 48, 49, 50, 52, 53, 56, 57, 58, 59, 61	3
7Networks_LH_Vis	1, 2, 3, 6, 7, 9, 12, 15, 18, 21, 24, 28, 29, 30, 33, 34, 36, 40, 42, 44, 45, 49, 53, 54, 58, 59, 61, 64, 65, 68, 70, 72, 74, 76, 77, 78, 79, 80, 81	3
7Networks_LH_Cont_PFC1	1, 2, 3, 4, 6, 7, 8, 9, 10, 11, 13, 14, 15, 16, 17, 18, 20	4
7Networks_LH_Cont_PFCv	1, 2	4
7Networks_LH_Cont_Temp	1	4
7Networks_LH_Default_PFC	1, 2, 3, 4, 6, 8, 11, 15, 17, 18, 21, 36, 42	4

Continued on next page

Table A.1 – continued

Parcel family	Parcel numbers	Cluster
7Networks_LH_Default_PHC	2	4
7Networks_LH_Default_Temp	1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 21	4
7Networks_LH_DorsAttn_FEF	1	4
7Networks_LH_DorsAttn_PrCv	1, 2, 3, 4	4
7Networks_LH_Limbic_OFC	1, 5	4
7Networks_LH_Limbic_TempPole	1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15	4
7Networks_LH_SalVentAttn_FrOperI ns	1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23	4
7Networks_LH_SalVentAttn_ParOper	3, 5, 7	4
7Networks_LH_SomMot	1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 20, 21, 22, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 43	4
7Networks_LH_Cont_Cing	2, 3, 5, 7, 8	5
7Networks_LH_Cont_pCun	4	5
7Networks_LH_Default_pCunPCC	11, 14, 16, 17, 19, 20, 21, 22, 23, 25, 26, 27, 28, 29, 31, 32	5
7Networks_LH_DorsAttn_Post	44, 45, 49	5
7Networks_LH_SalVentAttn_Med	5, 7, 8, 9, 11, 12, 13, 14, 16	5
7Networks_LH_SomMot	51, 54, 60, 62, 68, 73, 76, 78, 86, 87, 88, 91	5
7Networks_RH_Cont_Cing	1, 3, 5	5
7Networks_RH_Cont_pCun	2, 4	5
7Networks_RH_Default_pCunPCC	5, 8, 10, 11, 12, 13, 14, 16, 17, 18	5
7Networks_RH_DorsAttn_Post	33, 38, 41, 44, 45, 46, 47, 48, 49, 50, 51, 52	5

Continued on next page

Table A.1 – continued

Parcel family	Parcel numbers	Cluster
7Networks_RH_SalVentAttn_Med	6, 7, 8, 9, 10, 12, 13, 14, 16, 18	5
7Networks_RH_SomMot	51, 60, 66, 67, 72, 74, 76, 77, 79, 82, 84, 85, 86, 87, 88, 89, 90, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103	5
7Networks_RH_Cont_PFC1	20, 21, 23, 25, 26, 27, 29, 30, 31	6
7Networks_RH_Cont_Par	2, 3, 5, 6, 7, 8, 9, 10, 11, 12, 14, 15, 16, 17	6
7Networks_RH_Default_Par	11, 12, 14, 15	6
7Networks_RH_DorsAttn_FEF	1, 2, 3, 4, 5, 6	6
7Networks_RH_DorsAttn_Post	14, 17, 18, 20, 23, 25, 26, 28, 29, 30, 31, 34, 35, 36, 42	6
7Networks_RH_DorsAttn_PrCv	2, 3	6
7Networks_RH_SalVentAttn_Med	20	6
7Networks_RH_SalVentAttn_PrC	1, 2	6
7Networks_RH_SalVentAttn_TempOcc Par	12, 15, 16, 17, 18	6
7Networks_RH_SomMot	35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 46, 47, 48, 49, 50, 53, 54, 55, 56, 58, 59, 61, 62, 63, 65, 68, 69, 70, 71, 73, 75, 78, 80, 81	6
7Networks_LH_DorsAttn_FEF	4, 6	7
7Networks_LH_DorsAttn_Post	50	7
7Networks_LH_SomMot	63, 64, 66, 67, 69, 70, 71, 72, 74, 75, 77, 79, 80, 81, 83, 84, 85, 89, 90	7

A.2 Community Partition for SBMA

Table A.2: Parcel families and parcel numbers in each SBMA cluster.

Parcel family	Parcel numbers	Cluster
7Networks_LH_Cont_pCun	1, 2, 3	0
7Networks_LH_Default_PHC	3	0
7Networks_LH_Default_pCunPCC	1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 12, 13, 15, 18, 24, 28, 30	0
7Networks_LH_Vis	4, 5, 8, 10, 11, 13, 14, 16, 17, 19, 20, 22, 23, 25, 26, 27, 31, 32, 35, 37, 38, 39, 41, 43, 46, 47, 48, 50, 51, 52, 55, 56, 57, 60, 62, 63, 66, 67, 69, 71, 73, 75	0
7Networks_RH_Cont_Par	4, 13	0
7Networks_RH_Cont_pCun	1, 3, 5	0
7Networks_RH_Default_Par	16	0
7Networks_RH_Default_pCunPCC	1, 2, 3, 4, 6, 7, 9, 10, 15	0
7Networks_RH_DorsAttn_Post	2, 12, 13, 15, 16, 19, 21, 22, 24, 27, 32, 37, 39, 40, 43	0
7Networks_RH_Limbic_TempPole	16	0
7Networks_RH_SomMot	76	0
7Networks_RH_Vis	1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 22, 23, 24, 25, 27, 28, 29, 30, 31, 32, 33, 35, 36, 37, 38, 40, 41, 42, 43, 44, 45, 46, 47, 48, 50, 51, 52, 53, 54, 55, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81	0
7Networks_LH_Cont_Cing	1, 4, 6	1
7Networks_LH_Cont_OFC	1	1
7Networks_LH_Cont_PFCd	1	1
7Networks_LH_Cont_PFCl	5, 12, 19	1

Continued on next page

Table A.2 – continued

Parcel family	Parcel numbers	Cluster
7Networks_LH_Cont_PFCmp	1, 2	1
7Networks_LH_Default_PFC	5, 7, 9, 10, 12, 13, 14, 16, 19, 20, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 37, 38, 39, 40, 41, 43, 44, 45, 46, 47, 48, 49, 50	1
7Networks_LH_DorsAttn_FEF	2, 4, 5, 6, 7	1
7Networks_LH_Limbic_OFC	2, 3, 4, 6, 8, 9, 10, 11, 12, 13, 14	1
7Networks_LH_SalVentAttn_Med	1, 2, 3, 4, 6, 10, 15, 17, 18, 19	1
7Networks_LH_SalVentAttn_PFC1	1, 2	1
7Networks_LH_SomMot	42, 47, 55, 63, 65, 71, 75, 82, 85	1
7Networks_RH_Cont_Cing	2, 4	1
7Networks_RH_Cont_PFC1	1, 2, 3, 4, 5, 6, 8, 9, 11, 14, 17, 19, 22, 24, 28, 32, 33, 34	1
7Networks_RH_Cont_PFCmp	1, 2, 3, 4, 5	1
7Networks_RH_Default_PFCdPFCm	1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24	1
7Networks_RH_Limbic_OFC	2, 3, 4, 5, 7, 8, 9, 10, 11, 12, 13, 14	1
7Networks_RH_SalVentAttn_Med	1, 2, 3, 4, 5, 11, 15, 17, 19	1
7Networks_RH_SalVentAttn_PFC1	1, 2	1
7Networks_RH_SomMot	45, 52, 57, 64, 83, 91	1
7Networks_RH_Cont_PFC1	7, 10, 12, 13, 15, 16, 18, 20, 21	2
7Networks_RH_Cont_PFCv	1, 2	2
7Networks_RH_Cont_Par	1	2
7Networks_RH_Cont_Temp	1, 2, 3, 4	2
7Networks_RH_Default_PFCv	1, 2, 3, 4, 5, 6, 7, 8, 9, 10	2
7Networks_RH_Default_Par	1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 13	2

Continued on next page

Table A.2 – continued

Parcel family	Parcel numbers	Cluster
7Networks_RH_Default_Temp	1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18	2
7Networks_RH_DorsAttn_Post	1, 3, 4, 5, 6, 7, 8, 9, 10, 11	2
7Networks_RH_DorsAttn_PrCv	1, 2	2
7Networks_RH_Limbic_OFC	1, 6	2
7Networks_RH_Limbic_TempPole	1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 17	2
7Networks_RH_SalVentAttn_FrOperI ns	1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24	2
7Networks_RH_SalVentAttn_TempOcc Par	1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 13, 14	2
7Networks_RH_SomMot	1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34	2
7Networks_RH_Vis	21, 26, 34, 39, 49, 56	2
7Networks_LH_Cont_Par	1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15	3
7Networks_LH_Cont_Temp	2, 3, 4	3
7Networks_LH_Default_PHC	1	3
7Networks_LH_Default_Par	1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19	3
7Networks_LH_Default_Temp	20, 22	3
7Networks_LH_DorsAttn_FEF	3	3
7Networks_LH_DorsAttn_Post	1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 42, 43, 46, 47, 48	3
7Networks_LH_SalVentAttn_ParOper	1, 2, 4, 6, 8, 9	3

Continued on next page

Table A.2 – continued

Parcel family	Parcel numbers	Cluster
7Networks_LH_SalVentAttn_TempOcc	1, 2	3
7Networks_LH_SomMot	19, 23, 39, 40, 41, 44, 45, 46, 48, 49, 50, 52, 53, 56, 57, 58, 59, 61	3
7Networks_LH_Vis	1, 2, 3, 6, 7, 9, 12, 15, 18, 21, 24, 28, 29, 30, 33, 34, 36, 40, 42, 44, 45, 49, 53, 54, 58, 59, 61, 64, 65, 68, 70, 72, 74, 76, 77, 78, 79, 80, 81	3
7Networks_LH_Cont_PFC1	1, 2, 3, 4, 6, 7, 8, 9, 10, 11, 13, 14, 15, 16, 17, 18, 20	4
7Networks_LH_Cont_PFCv	1, 2	4
7Networks_LH_Cont_Temp	1	4
7Networks_LH_Default_PFC	1, 2, 3, 4, 6, 8, 11, 15, 17, 18, 21, 36, 42	4
7Networks_LH_Default_PHC	2	4
7Networks_LH_Default_Temp	1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 21	4
7Networks_LH_DorsAttn_FEF	1	4
7Networks_LH_DorsAttn_PrCv	1, 2, 3, 4	4
7Networks_LH_Limbic_OFC	1, 5, 7	4
7Networks_LH_Limbic_TempPole	1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15	4
7Networks_LH_SalVentAttn_FrOperIns	1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23	4
7Networks_LH_SalVentAttn_ParOper	3, 5, 7	4
7Networks_LH_SomMot	1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 20, 21, 22, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 43	4

Continued on next page

Table A.2 – continued

Parcel family	Parcel numbers	Cluster
7Networks_LH_Cont_Cing	2, 3, 5, 7, 8	5
7Networks_LH_Cont_pCun	4	5
7Networks_LH_Default_pCunPCC	11, 14, 16, 17, 19, 20, 21, 22, 23, 25, 26, 27, 29, 31, 32	5
7Networks_LH_DorsAttn_Post	41, 44, 45, 49, 50	5
7Networks_LH_SalVentAttn_Med	5, 7, 8, 9, 11, 12, 13, 14, 16	5
7Networks_LH_SomMot	51, 54, 60, 62, 64, 66, 67, 68, 69, 70, 72, 73, 74, 76, 77, 78, 79, 80, 81, 83, 84, 86, 87, 88, 89, 90, 91	5
7Networks_RH_Cont_Cing	1, 3, 5	5
7Networks_RH_Cont_pCun	2, 4	5
7Networks_RH_Default_pCunPCC	5, 8, 11, 12, 13, 14, 16, 17, 18	5
7Networks_RH_DorsAttn_Post	33, 38, 41, 44, 45, 46, 47, 48, 49, 50, 51, 52	5
7Networks_RH_SalVentAttn_Med	6, 7, 8, 9, 10, 12, 13, 14, 16, 18	5
7Networks_RH_SomMot	51, 60, 66, 67, 72, 74, 77, 79, 82, 84, 85, 86, 87, 88, 89, 90, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103	5
7Networks_RH_Cont_PFC1	23, 25, 26, 27, 29, 30, 31	6
7Networks_RH_Cont_Par	2, 3, 5, 6, 7, 8, 9, 10, 11, 12, 14, 15, 16, 17	6
7Networks_RH_Default_Par	11, 12, 14, 15	6
7Networks_RH_DorsAttn_FEF	1, 2, 3, 4, 5, 6	6
7Networks_RH_DorsAttn_Post	14, 17, 18, 20, 23, 25, 26, 28, 29, 30, 31, 34, 35, 36, 42	6
7Networks_RH_DorsAttn_PrCv	3	6
7Networks_RH_SalVentAttn_PrC	1, 2	6
7Networks_RH_SalVentAttn_TempOcc Par	12, 15, 16, 17, 18	6

Continued on next page

Table A.2 – continued

Parcel family	Parcel numbers	Cluster
7Networks_RH_SomMot	35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 46, 47, 48, 49, 50, 53, 54, 55, 56, 58, 59, 61, 62, 63, 65, 68, 69, 70, 71, 73, 75, 78, 80, 81	6

Bibliography

- [1] C. Bertolin, G. Querin, I. Martinelli, M. Pennuto, E. Pegoraro, and G. Sorarù, “Insights into the genetic epidemiology of spinal and bulbar muscular atrophy: Prevalence estimation and multiple founder haplotypes in the Veneto Italian region,” *European Journal of Neurology*, vol. 26, no. 3, pp. 519–524, 2019.
- [2] A. P. Lieberman, “Chapter 40 - Spinal and bulbar muscular atrophy,” in *Neurogenetics, Part II* (D. H. Geschwind, H. L. Paulson, and C. Klein, eds.), vol. 148 of *Handbook of Clinical Neurology*, pp. 625–632, Elsevier, 2018.
- [3] P.-F. Pradat, “[sbma: a rare disease but a classic als mimic syndrome],” *Presse Medicale*, vol. 43, no. 5, pp. 580–586, 2014.
- [4] A. La Spada, “Spinal and Bulbar Muscular Atrophy,” in *GeneReviews®* (M. P. Adam, S. Bick, G. M. Mirzaa, S. E. Wallace, and A. Amemiya, eds.), University of Washington, Seattle, 1993.
- [5] G. Kuhlenbäumer, W. Kress, E. Ringelstein, and F. Stögbauer, “Thirty-seven CAG repeats in the androgen receptor gene in two healthy individuals,” *Journal of Neurology*, vol. 248, no. 1, pp. 23–26, 2001.
- [6] P.-F. Pradat, E. Bernard, P. Corcia, P. Couratier, C. Jublanc, G. Querin, C. Morélot Panzini, F. Salachas, C. Vial, K. Wahbi, P. Bede, C. Desnuelle, N. Le Forestier, A. Echaniz-Laguna, G. Querin, G. Sorarù, T. Perez, C. Ramos, C. Goizet, J. C. Desport, M. Pugeat, B. Pichon, S. Maniez, J. Robillard, C. Coupe, L. Laurier Betram, S. Roy Bellina, N. Lévêque, J. Penot, V. Goutines Caramel, and on behalf of the French Kennedy’s Disease Writing Group, “The French national protocol for Kennedy’s disease (SBMA): Consensus diagnostic and management recommendations,” *Orphanet Journal of Rare Diseases*, vol. 15, no. 1, p. 90, 2020.
- [7] H. Adachi, M. Waza, M. Katsuno, F. Tanaka, M. Doyu, and G. Sobue, “Pathogenesis and molecular targeted therapy of spinal and bulbar muscular atrophy,” *Neuropathology and Applied Neurobiology*, vol. 33, no. 2, pp. 135–151, 2007.

- [8] W. R. Kennedy, M. Alter, and J. H. Sung, "Progressive proximal spinal and bulbar muscular atrophy of late onset: A sex-linked recessive trait," *Neurology*, vol. 18, no. 7, pp. 671–671, 1968.
- [9] F. J. Arnold and D. E. Merry, "Molecular Mechanisms and Therapeutics for SBMA/Kennedy's Disease," *Neurotherapeutics*, vol. 16, no. 4, pp. 928–947, 2019.
- [10] R. Manzano, G. Sorarú, C. Grunseich, P. Fratta, E. Zuccaro, M. Pennuto, and C. Rinaldi, "Beyond motor neurons: Expanding the clinical spectrum in Kennedy's disease," *Journal of Neurology, Neurosurgery & Psychiatry*, vol. 89, no. 8, pp. 808–812, 2018.
- [11] A. Hashizume, K. H. Fischbeck, M. Pennuto, P. Fratta, and M. Katsuno, "Disease mechanism, biomarker and therapeutics for spinal and bulbar muscular atrophy (SBMA)," *Journal of Neurology, Neurosurgery & Psychiatry*, vol. 91, no. 10, pp. 1085–1091, 2020.
- [12] H. Nakatsuji, A. Araki, A. Hashizume, Y. Hijikata, S. Yamada, T. Inagaki, K. Suzuki, H. Banno, N. Suga, Y. Okada, M. Ohyama, T. Nakagawa, K. Kishida, T. Funahashi, I. Shimomura, H. Okano, M. Katsuno, and G. Sobue, "Correlation of insulin resistance and motor function in spinal and bulbar muscular atrophy," *Journal of Neurology*, vol. 264, no. 5, pp. 839–847, 2017.
- [13] N. Atsuta, H. Watanabe, M. Ito, H. Banno, K. Suzuki, M. Katsuno, F. Tanaka, A. Tamakoshi, and G. Sobue, "Natural history of spinal and bulbar muscular atrophy (SBMA): A study of 223 Japanese patients," *Brain*, vol. 129, no. 6, pp. 1446–1455, 2006.
- [14] S. Tu, T. Li, A. S. Carroll, C. J. Mahoney, W. Huynh, S. B. Park, R. Henderson, S. Vucic, M. C. Kiernan, and C. S.-Y. Lin, "Central neurodegeneration in Kennedy's disease accompanies peripheral motor dysfunction," *Scientific Reports*, vol. 14, no. 1, p. 18331, 2024.
- [15] T.-H. Lai, R.-S. Liu, B.-H. Yang, P.-S. Wang, K.-P. Lin, Y.-C. Lee, and B.-W. Soong, "Cerebral involvement in spinal and bulbar muscular atrophy (Kennedy's disease): A pilot study of PET," *Journal of the Neurological Sciences*, vol. 335, no. 1, pp. 139–144, 2013.
- [16] J. Kassubek, F. D. Juengling, and A. Sperfeld, "Widespread white matter changes in Kennedy disease: A voxel based morphometry study," *Journal of Neurology, Neurosurgery, and Psychiatry*, vol. 78, no. 11, pp. 1209–1212, 2007.

- [17] G. De Luca, E. N. Aiello, A. Moreschi, B. Curti, S. Marcato, F. Paredi, F. Verde, V. Silani, N. Ticozzi, B. Poletti, and G. Sorarù, “Cognitive and behavioral involvement in spinal and bulbar muscular atrophy (SBMA): A systematic review,” *Amyotrophic Lateral Sclerosis and Frontotemporal Degeneration*, vol. 0, no. 0, pp. 1–19, 2026.
- [18] T. Hastie, R. Tibshirani, and J. Friedman, *The Elements of Statistical Learning: Data Mining, Inference, and Prediction*. Springer Series in Statistics, Springer, 2 ed., 2009.
- [19] P. A. Lachenbruch and M. R. Mickey, “Estimation of Error Rates in Discriminant Analysis,” *Technometrics*, vol. 10, no. 1, pp. 1–11, 1968.
- [20] A. F. McDaid, D. Greene, and N. Hurley, “Normalized Mutual Information to evaluate overlapping community finding algorithms.” arXiv preprint arXiv:1110.2515, 2013.
- [21] M. Sundqvist, J. Chiquet, and G. Rigai, “Adjusting the adjusted Rand Index – A multinomial story.” arXiv preprint arXiv:2011.08708, 2020.
- [22] R. Real and J. M. Vargas, “The Probabilistic Basis of Jaccard’s Index of Similarity,” *Systematic Biology*, vol. 45, no. 3, pp. 380–385, 1996.
- [23] M. Sokolova and G. Lapalme, “A systematic analysis of performance measures for classification tasks,” *Information Processing & Management*, vol. 45, no. 4, pp. 427–437, 2009.
- [24] A.-L. Barabási and M. Pósfai, *Network Science*. Cambridge University Press, 2016.
- [25] E. Cozzo *et al.*, *Multiplex Networks: Basic Formalism and Structural Properties*. Springer, 2018.
- [26] S. Fortunato, “Community detection in graphs,” *Physics Reports*, vol. 486, no. 3–5, pp. 75–174, 2010.
- [27] P. J. Mucha, T. Richardson, K. Macon, M. A. Porter, and J.-P. Onnela, “Community Structure in Time-Dependent, Multiscale, and Multiplex Networks,” *Science*, vol. 328, no. 5980, pp. 876–878, 2010.
- [28] V. D. Blondel, J.-L. Guillaume, R. Lambiotte, and E. Lefebvre, “Fast unfolding of communities in large networks,” *Journal of Statistical Mechanics: Theory and Experiment*, vol. 2008, no. 10, p. P10008, 2008.

- [29] M. E. J. Newman, “Modularity and community structure in networks,” *Proceedings of the National Academy of Sciences*, vol. 103, no. 23, pp. 8577–8582, 2006.
- [30] V. A. Traag, L. Waltman, and N. J. van Eck, “From Louvain to Leiden: Guaranteeing well-connected communities,” *Scientific Reports*, vol. 9, no. 1, p. 5233, 2019.
- [31] S. Monti, P. Tamayo, J. Mesirov, and T. Golub, “Consensus Clustering: A Resampling-Based Method for Class Discovery and Visualization of Gene Expression Microarray Data,” *Machine Learning*, vol. 52, no. 1, pp. 91–118, 2003.
- [32] B. C. Ross, “Mutual information between discrete and continuous data sets,” *PLoS ONE*, vol. 9, no. 2, p. e87357, 2014.
- [33] J. M. Moguerza and A. Muñoz, “Support Vector Machines with Applications,” *Statistical Science*, vol. 21, no. 3, 2006.
- [34] S. Sperandei, “Understanding logistic regression analysis,” *Biochemia Medica*, vol. 24, no. 1, pp. 12–18, 2014.
- [35] L. Breiman, J. Friedman, R. A. Olshen, and C. J. Stone, *Classification and Regression Trees*. Chapman and Hall/CRC, 2017.
- [36] G. Louppe, “Understanding Random Forests: From Theory to Practice.” arXiv preprint arXiv:1407.7502, 2015.
- [37] H.-H. Chen, “Understanding Gradient Boosting Classifier: Training, Prediction, and the Role of γ_j .” arXiv preprint arXiv:2410.05623, 2024.
- [38] T. Chen and C. Guestrin, “XGBoost: A Scalable Tree Boosting System,” in *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pp. 785–794, 2016.
- [39] W. Hsieh, Z. Bi, C. Jiang, J. Liu, B. Peng, S. Zhang, X. Pan, J. Xu, J. Wang, K. Chen, P. Feng, Y. Wen, X. Song, T. Wang, M. Liu, J. Yang, M. Li, B. Jing, J. Ren, J. Song, H.-M. Tseng, Y. Zhang, L. K. Q. Yan, Q. Niu, S. Chen, Y. Wang, and C. X. Liang, “A Comprehensive Guide to Explainable AI: From Classical Models to LLMs.” arXiv preprint arXiv:2412.00800, 2024.
- [40] S. Lundberg and S.-I. Lee, “A Unified Approach to Interpreting Model Predictions.” arXiv preprint arXiv:1705.07874, 2017.

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- [41] L. S. Shapley, “A value for n-person games,” in *Contributions to the Theory of Games II*, Princeton University Press, 1953.
 - [42] A. Schaefer, R. Kong, E. M. Gordon, T. O. Laumann, X.-N. Zuo, A. J. Holmes, S. B. Eickhoff, and B. T. T. Yeo, “Local-Global Parcellation of the Human Cerebral Cortex from Intrinsic Functional Connectivity MRI,” *Cerebral Cortex*, vol. 28, no. 9, pp. 3095–3114, 2018.
 - [43] B. T. T. Yeo, F. M. Krienen, J. Sepulcre, M. R. Sabuncu, D. Lashkari, M. Hollinshead, J. L. Roffman, J. W. Smoller, L. Zöllei, J. R. Polimeni, B. Fischl, H. Liu, and R. L. Buckner, “The organization of the human cerebral cortex estimated by intrinsic functional connectivity,” *Journal of Neurophysiology*, vol. 106, no. 3, pp. 1125–1165, 2011.